

ACHIEVE!



Science understanding

CHEMISTRY FOR YEARS 7–8

Janine Haeusler



Made for the
Australian Curriculum

The periodic table

1 H Hydrogen		Key → non-metals → metals → unknown properties																2 He Helium																	
3 Li Lithium		4 Be Beryllium																		9 F Fluorine		10 Ne Neon													
11 Na Sodium		12 Mg Magnesium																		17 Cl Chlorine		18 Ar Argon													
19 K Potassium		20 Ca Calcium		21 Sc Scandium		22 Ti Titanium		23 V Vanadium		24 Cr Chromium		25 Mn Manganese		26 Fe Iron		27 Co Cobalt		28 Ni Nickel		29 Cu Copper		30 Zn Zinc		31 Ga Gallium		32 Ge Germanium		33 As Arsenic		34 Se Selenium		35 Br Bromine		36 Kr Krypton	
37 Rb Rubidium		38 Sr Strontium		39 Y Yttrium		40 Zr Zirconium		41 Nb Niobium		42 Mo Molybdenum		43 Tc Technetium		44 Ru Ruthenium		45 Rh Rhodium		46 Pd Palladium		47 Ag Silver		48 Cd Cadmium		49 In Indium		50 Sn Tin		51 Sb Antimony		52 Te Tellurium		53 I Iodine		54 Xe Xenon	
55 Cs Caesium		56 Ba Barium																		81 Tl Thallium		82 Pb Lead		83 Bi Bismuth		84 Po Polonium		85 At Astatine		86 Rn Radon					
87 Fr Francium		88 Ra Radium		104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium		111 Rg Roentgenium		112 Uub Ununbium		113 Uut Ununtrium		114 Uuq Ununquadium		115 Uup Ununpentium		116 Uuh Ununhexium		117 Uus Ununseptium		118 Uuo Ununoctium			

57	La		58	Ce		59	Pr		60	Nd		61	Pm		62	Sm		63	Eu		64	Gd		65	Tb		66	Dy		67	Ho		68	Er		69	Tm		70	Yb		71	Lu
Lanthanum			Cerium			Praseodymium			Neodymium			Promethium			Samarium			Europium			Gadolinium			Terbium			Dysprosium			Holmium			Erbium			Thulium			Ytterbium			Lutetium	
89	Ac		90	Th		91	Pa		92	U		93	Np		94	Pu		95	Am		96	Cm		97	Bk		98	Cf		99	Es		100	Fm		101	Md		102	No		103	Lr
Actinium			Thorium			Protactinium			Uranium			Neptunium			Plutonium			Americium			Curium			Berkellium			Californium			Einsteinium			Fermium			Mendelevium			Nobelium			Lawrencium	



ACHIEVE!

Science understanding

CHEMISTRY FOR YEARS 7–8

Janine Haeusler

This book was commissioned by Lisa Tancredi

Editing and project management by Vanessa Lanaway

Typesetting by digiXL

Cover design by Jo-Anne Ridgway/Polar Design

.....

Janine Haeusler has worked as an educational consultant for five years, reviewing and writing educational resources for science and mathematics teachers. She has more than 12 years' experience as a science and mathematics teacher in rural Western Australia, and has also worked as a Student At Educational Risk Teacher, helping teachers develop appropriate programs and resources for struggling students. Having been educated in rural schools herself, she is passionate about rural education and the need for specialised and highly experienced teachers in rural schools. Janine holds a Bachelor and a Masters of Education, and has written titles in the *Achieve! Interactive Instant Lessons* and *Achieve! Science Understanding* series for Blake Education.

© Janine Haeusler and Blake Education 2014

Blake Education owns all copyrights in the literary and the artistic works in this book.

All material identified by  is material subject to copyright under the *Copyright Act 1968* (Cth) and is owned by the Australian Curriculum, Assessment and Reporting Authority 2013.

For all Australian Curriculum material except elaborations: This is an extract from the Australian Curriculum.

Elaborations: This may be a modified extract from the Australian Curriculum and may include the work of other authors.

Disclaimer: ACARA neither endorses nor verifies the accuracy of the information provided and accepts no responsibility for incomplete or inaccurate information. In particular, ACARA does not endorse or verify that:

- The content descriptions are solely for a particular year and subject;
- All the content descriptions for that year and subject have been used; and
- The author's material aligns with the Australian Curriculum content descriptions for the relevant year and subject.

You can find the unaltered and most up to date version of the material at <http://www.australiancurriculum.edu.au>

This material is reproduced with the permission of ACARA.

Copying of this book for educational purposes

The purchasing educational institution may only photocopy pages within this book in accordance with the *Australian Copyright Act 1968* (the Act) and provided the educational institution (or body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act. It is a breach of copyright to scan or in any other way make digital copies of the work sheets.

It is mandatory that ALL photocopies are recorded by the institution for CAL survey purposes.

For details of the CAL licence for educational institutions contact:

Copyright Agency Limited
Level 15, 233 Castlereagh Street
Sydney NSW 2000
Telephone: (02) 9394 7600
Facsimile: (02) 9394 7601
E-mail: enquiry@copyright.com.au

The purchasing educational institution may store this book on a retrieval system only in accordance with the terms of the User Agreement printed on the inside back cover of this title.

Printed in Singapore by Green Giant Press

Blake
EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
info@blake.com.au
www.blake.com.au

ISBN 978 1 92571 061 8

Disclaimer: Every effort has been made to present activities in a safe way, warning teachers of potential hazards that can be foreseen. However, because requirements differ between states and often change over time, it is each teacher's responsibility to do their own risk assessment for any activity they choose to do with their classes.

The publisher and the author have made every reasonable effort to ensure that the experiments and activities in this book are safe when conducted as instructed, but assume no responsibility for any damage caused or sustained. Students should carry out all experiments under close supervision.

Every effort has been made to contact and acknowledge the copyright holders of material used in this publication. If any copyright holder has been overlooked, we would be pleased to make the necessary arrangements to address the situation.

Introduction	4		
1 Matter	5	6 Compounds ▲	35
What is matter?	6	Joining elements (1)	36
Measuring matter	7	Joining elements (2)	37
Materials around us	8	Common compounds	38
Properties of materials	9	Drawing and making molecules	39
Materials and their uses	10	Writing chemical formulas	40
2 Solids, liquids and gases ▲	11	7 Mixtures and solutions ▲	41
The particle model of matter (1)	12	Mixtures and pure substances	42
The particle model of matter (2)	13	Solutions	43
Viscosity of liquids (1) ▲	14	Investigating solvents (1) ▲	44
Viscosity of liquids (2) ▲	15	Investigating solvents (2) ▲	45
What is diffusion?	16	Factors affecting solubility	46
3 Changes of state ▲	17	8 Separating mixtures ▲	47
Heating up	18	Filtration	48
Cooling down	19	Filters in and around the home	49
Boiling and melting points	20	Evaporation and crystallisation	50
Temperature and state (1)	21	Distillation (1)	51
Temperature and state (2)	22	Distillation (2)	52
4 Atoms and elements ▲	23	9 Physical and chemical changes ▲	53
Atoms and elements	24	Changing matter	54
The periodic table (1)	25	Physical and chemical changes	55
The periodic table (2)	26	Is it a chemical change? ▲	56
Naming elements	27	Evidence of a chemical change	57
Elements in the human body	28	Precipitates ▲	58
5 Metals and non-metals ▲	29	10 Chemical reactions ▲	59
Metals and non-metals	30	Reactants and products	60
Properties of metals	31	Writing word equations	61
Properties of non-metals	32	Rusting	62
Carbon	33	Burning	63
Reactive metals ▲	34	Chemical reactions in nature	64

The *Achieve!* series of science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

Chemistry for years 7–8 will assist teachers to develop lessons for the chemistry sub-strand in *Science understanding* for years 7 and 8. Units of work cover a variety of topics, including the different states of matter, elements, compounds and mixtures, solutions and a range of separation techniques, and physical and chemical changes. Students build their scientific vocabulary and understanding of key concepts. Work sheets involving practical activities encourage students to plan and safely conduct experiments and to record and analyse data. Practical activities are clearly marked with the work safely symbol ▲.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the individual needs of students.

This book is divided into ten units, containing photocopiable Resource and Activity sheets. This will allow teachers to use the material in a variety of ways. For example, the unit could be taught as a lesson with students in groups of varying sizes. Alternatively, a single Resource sheet and related Activity sheets can be used as support material, or for individual homework, assessment or revision.

The teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the work sheets.

Prior knowledge – This refers to the skills and knowledge required for students to complete the tasks. Some activities are more challenging than others, however, as a rule, students should have a reading comprehension age of six to nine years to be able to attempt the activities.

Background – This section provides additional guidance to help teachers to present a successful lesson. It may include definitions or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Resource and Activity work sheets – The Resource sheets do not always contain activities and are used either to stimulate discussion or as a part of the unit's activities. The Activity sheets may be used sequentially or as stand-alone tasks, depending on the needs of the students.

Extension ideas – Additional tasks are suggested to recap on the main skills or knowledge taught in the lesson, as well as ways to extend, assess or include technology in the lesson.

The Australian Curriculum alignment		Unit									
	Year level	1	2	3	4	5	6	7	8	9	10
Mixtures, including solutions, contain a combination of pure substances that can be separated using a range of techniques (ACSSU113)	7							✓	✓		
The properties of the different states of matter can be explained in terms of the motion and arrangement of particles (ACSSU151)	8	✓	✓	✓							
Differences between elements, compounds and mixtures can be described at a particle level (ACSSU152)	8				✓	✓	✓				
Chemical change involves substances reacting to form new substances (ACSSU225)	8									✓	✓
People use understanding and skills from across the disciplines of science in their occupations (ACSHE227)	8	✓									

Objectives

- To understand the difference between forms of energy and matter
- To understand that matter can be measured
- To know the difference between natural and processed materials
- To recognise the properties of different materials, including metals, plastics and ceramics
- To identify the different materials used for different objects and relate to their properties

Prior knowledge

Students should recognise that everything they can touch and see is made up of matter. Even gases, which often cannot be seen, are made of matter.

Background

Matter is made up of tiny particles called atoms. There are millions of different materials, which scientists classify into different groups such as metals, plastics and ceramics. Materials can be natural or processed. The properties of a material determine its function.

Starter activity

Ask students to list as many of the objects that they can see around them in one minute. Have them group their objects according to the different materials they are made from. Finally, ask students to group them again, this time based on whether they think they are made from natural (raw) or processed (man-made) materials.

Activity sheets

The Activity sheet 'What is matter?' explains the difference between matter and energy and introduces some characteristics of matter. Students complete a cloze passage and then sort the examples provided into either matter or forms of energy. They then classify different states of water as either solid, liquid or gas.

The 'Measuring matter' Activity sheet describes some of the ways we can measure matter. Students read a passage and identify the different ways measurements can be taken and then identify the correct scientific equipment used to do this.

The Activity sheet 'Materials around us' explains the difference between natural and processed materials. Students complete a passage about natural and processed materials by filling in the missing vowels, before identifying examples as either natural or processed. There are also phrases to match.

The 'Properties of materials' Activity sheet introduces some of the different properties that can be used to describe different materials. Students match different properties to their description. Using the notes about plastics, students should be able to expand the information into a paragraph.

The 'Materials and their uses' Activity sheet relates the properties of different materials to their uses in various objects. Students read a passage about different objects and the materials they are made from. They then use this information to complete the table and explain why each particular material is used. To get students thinking about scientific inventions and discoveries, they are asked to describe an object that was not available when their parent/s or guardian/s were children and to list the materials it is made from.

Extension idea

- Ask students to research different smart materials. They could make a poster detailing their chosen smart material and demonstrating potential everyday uses.

Selected answers

What is matter? 2 Matter: car, iPad, petrol, oxygen; Energy: electricity, thunder, heat, lightning

Measuring matter 2 (left to right) spring balance: weight; measuring cylinder: volume; triple beam balance: mass.

What is matter?

 1 Use the words in the word bank to complete the passage below.

Everything in the world belongs to two groups: e_____ or m_____. Energy makes things happen. Different forms of energy include l_____, s_____ and h_____. Matter is anything that can be w_____ and that takes up s_____. All material objects are made up of matter.

Living o_____ like plants and a_____ are also made up of matter.

Matter can be either a s_____, l_____ or g_____. These are known as the three s_____ of matter. Water can exist in all three of these states.

2 Decide which of the following are matter and which are forms of energy. Write them in the table below.

Word bank

light
solid
matter
heat
gas
energy
sound
organisms
states
space
liquid
weighed
animals

car	electricity	iPad	thunder
petrol	heat	oxygen	lightning

Matter	Energy



3 Water can be a solid, a liquid or a gas. Match each form of water to the correct state.

solid	water vapour
liquid	ice
gas	liquid water

Measuring matter

Matter takes up space. This means that it has **volume**. Volume is one measurement of matter. You can use a formula to calculate the volume of regular-shaped objects. You can also measure the volume of an object by dropping it in water and calculating the amount of water it **displaces**, or pushes aside.

Mass is another measurement of matter. Mass describes the amount of matter or **atoms** in a substance. Matter can be measured by finding the **weight** of an object. An object's weight depends on its mass and how much **gravity** is pulling on it. For example, an object weighs less on the moon than on earth because there is less gravity on the moon. We can use a spring balance to measure weight.

The **density** of an object depends on both its mass and its volume. Objects with tightly packed atoms are very dense. Objects with atoms spaced further apart are less dense. Density is measured in grams per cubic centimetre (g/cm^3) or kilograms per cubic metre (kg/m^3).

 **1** Read the information above. Underline the ways matter can be measured.

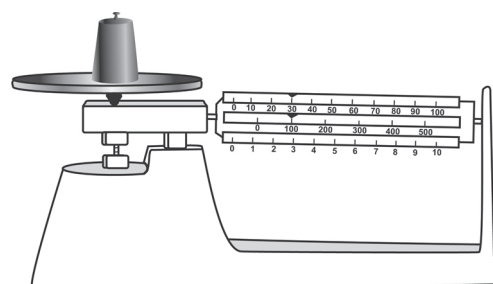
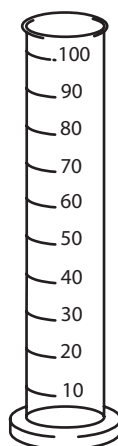
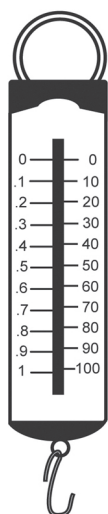
2 Many different instruments can be used to measure matter. Some are shown below. Use the words in the word bank to label each piece of equipment, then write down what each instrument could be used to measure.

Word bank

Spring balance

Triple beam balance

Measuring cylinder



Materials around us

 1 Fill in the missing vowels (a, e, i, o, u) to complete the following passage.

M__t__r__ __ls are the th__ngs that objects are m__d__ from. M__t__r__ __ls can be m__n-m__d__ or n__t__r__l. Natural materials or r__s__ __rc__s are anything that comes from n__t__r__ that people can use, such as w__t__r, w __ __d, pl__nts, o __l and c__ __l. Natural materials are also called r__w materials.

Pr__c__ss__d materials are man-made using one or more n__t__r__l materials. For example, pl__st__cs are made from crude o __l. Gl__ss is a man-made material made from s__ nd, l__m__ and other minerals. A number of different raw materials, including c__rb__n, __r __n, chromium and nickel, are added together to make st__ __ nl__ss st__ __ l.

2 Decide which of the materials listed below are natural and which are processed (man-made). Write them under the correct headings.

tree	sand	plastic	paint	corn	steel
water	wheat	flour	glass	gold	cement

Natural materials	Processed materials (man-made)

3 Match the phrases below.

- Concrete is made from
- iron, chromium, carbon and nickel.
- Plastic is made from
- blue metal, sand, cement and water.
- Stainless steel is made from
- sand, lime and other minerals.
- Glass is made from
- crude oil.

Properties of materials

 1 Read the passage below and underline the correctly spelled word in each pair.

The word (**properties, propertees**) is used to describe the characteristics of a particular material. Different materials have different properties. There are lots of different words that can be used to describe the properties of an object. A (**flexable/flexible**) object is able to bend without breaking. Another object may be (**ridged/rigid**), which is the opposite of flexible. If you cannot see through an object, it is (**opaque/opake**). Objects that you can see through, such as windows, are **transparent**. Rubber bands are **elastic** because they can (**stretch/strech**) and then return to their original shape. **Absorbency** is a property of sponges – that’s why we use them to clean up spills. Metals are good **conductors** because they allow heat and (**electrisitee/electricity**) to pass through them.

2 Match each property with its description.

absorbent	does not change shape easily
good insulator	stretches then returns to original shape
transparent	can be seen through
elastic	soaks up liquids
rigid	does not allow heat or electricity to pass through easily
good conductor	cannot be seen through
opaque	allows electricity or heat to flow through it

3 Use the notes below to write a paragraph about the properties of plastic.

- | | |
|--------------------------------|--|
| • does not conduct electricity | • used to insulate electrical cords |
| • strong and lightweight | • can be moulded into different shapes |
| • does not corrode easily | • good insulator of heat |

Materials and their uses

The ways a material can be used depend on its properties.

- A saucepan is usually made out of metal and plastic. Metal is used because it is a good conductor of heat. Plastic is used for the handle because plastic is an **insulator**, or poor conductor of heat. This stops you burning your hand when you pick up the saucepan.
- Electrical cables are often made out of a metal called copper because it is a good conductor of electricity and can easily be formed into wires. The copper wire is surrounded with a plastic covering so the electrical cable can be handled safely.
- Ceramic materials are used for lots of different things. Traditional ceramics are used to make dishes and tiles because they are hard, do not absorb liquid and are both heat and rust resistant. Space shuttles have tiles made from ceramics because they protect the shuttle from extreme temperatures.

Scientists are always trying to design better or smarter materials to develop new technology and improve the way objects work. Smart materials are able to respond to changes in the environment.

 1 Complete the table below.

Object	Materials used	Reason why these materials are used
dinner plate		
saucepan		
space shuttle		
electrical cable		

- 2 Name one piece of equipment you use that your parent/s or guardian/s never had as a child. Describe what it is made from and what it is used for.

Object: _____

Objectives

- To understand that the properties of the different states of matter can be explained in terms of motion and arrangement of particles
- To identify simple particle models for each of the different states of matter
- To realise that particles in matter are always moving
- To understand that because the particles in liquids and gases can move about, they diffuse
- To make careful and accurate observations and display data in the form of a table and a graph
- To draw conclusions from experimental data

Prior knowledge

Students should be able to classify matter as a solid, liquid or gas.

Background

The particle model says that matter is made up of tiny units called particles. In solids, the particles are regularly arranged and touching and can only vibrate. In liquids, the particles are also touching but they are randomly arranged, allowing them to roll or slide over each other. The particles in gases are randomly arranged and far apart. They move about quickly and in all directions. The different properties of solids, liquids and gases can be explained in terms of motion and arrangement of particles.

Starter activity

Students should understand that solids, liquids and gases behave differently. Show students an animated video on solids, liquids and gases, for example www.youtube.com/watch?v=btGu9FWSPtc, then ask them to brainstorm what they know about the different states of matter.

Resource and Activity sheets

The Resource sheet 'The particle model of matter (1)' explains the differences between the three states of matter in terms of particle arrangement and motion. It identifies the different characteristics of solids, liquids and gases, explaining how these properties are due to particle arrangement and motion. Students should refer to this page when completing the Activity sheet 'The particle model of matter (2)', which requires

students to draw the arrangement of particles in the three states of matter and identify the characteristic of solids, liquids and gases. Some true/false statements conclude the activity.

The 'Viscosity of liquids' (1) and (2) Activity sheets develop students' skills in conducting experiments and recording and analysing data, while investigating the property of viscosity in different liquids. Each group of students will need a stopwatch, a marble, a 100-mL measuring cylinder and five different liquids. Make sure students drop the marble in gently to avoid breaking the measuring cylinder.

The 'What is diffusion?' Activity sheet introduces students to the concept of diffusion of liquids and gases. Students read a passage explaining diffusion, and choose the correct word from each of the paired options. They are then asked to list three examples of diffusion. Students complete the Activity sheet by choosing the correct endings for four different statements about diffusion.

Extension ideas

- Students can play an online game about solids, liquids and gases. There are numerous games available on the internet.
- Students could perform an experiment to observe diffusion.



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

The particle model of matter (1)

There are three states of matter: solid, liquid and gas. The **particle model** is a scientific explanation that describes the differences between each state. When we imagine particles in the three states of matter, it is as if we are looking at objects with a very powerful microscope and seeing what is happening inside them.

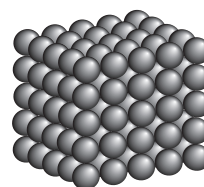
Solids

The particles in solids:

- are very close together
- are tightly held together
- have a small amount of energy
- can only move by vibrating.

Because of this, solids:

- are heavy
- have a definite shape
- do not flow easily
- are hard to compress.



The arrangement of particles in solids

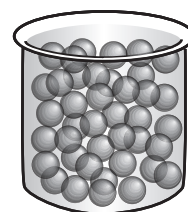
Liquids

The particles in liquids:

- are close together
- are weakly held together
- have a larger amount of energy
- vibrate, but can also slide over each other and change places.

Because of this, liquids:

- are heavy
- can be poured or flow freely
- have no definite shape
- fill the shape of a container
- can't be compressed.



The arrangement of particles in liquids

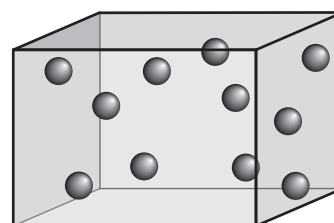
Gases

The particles in gases:

- are spread far apart
- are free from each other
- have a large amount of energy
- move around quickly and freely in all directions.

Because of this, gases:

- are light
- can change shape
- flow freely
- do not have a definite shape or volume
- spread out to fill a container
- can be compressed.

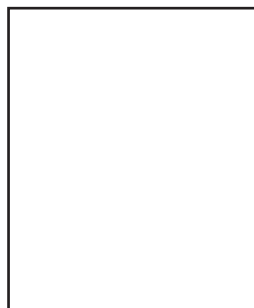


The arrangement of particles in gases

The particle model of matter (2)

 Read the Resource sheet 'The particle model of matter (1)', then answer these questions.

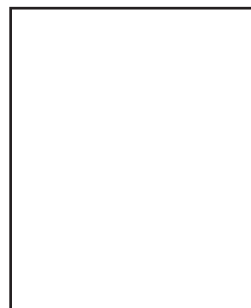
1 In the boxes below, show how particles are arranged in each state of matter.



Solids



Liquids



Gases

2 Use the phrases in the word bank to complete the table below. Some phrases may be used more than once.

Word bank

slide over each other flow freely are heavy fill the shape of a container
are light have a definite shape can change shape don't flow easily
only vibrate move quickly can't be compressed

The particles in solids ...	The particles in liquids ...	The particles in gases ...
Because of this, solids ...	Because of this, liquids ...	Because of this, gases ...

3 Decide whether each of these statements is true or false.

a Liquids have a fixed shape. _____ c Particles in solids can move freely. _____

b Gases are light. _____ d Liquids flow freely. _____

Viscosity of liquids (1)



One property of liquids is **viscosity**, which describes a liquid's resistance to flowing. Liquids such as water and milk are quite thin and flow more easily than thicker, more **viscous** liquids, such as honey.

A simple way to observe the viscosity of a liquid is to drop a marble into a cylinder filled with the liquid and measure how long the marble takes to sink to the bottom. The longer it takes to move through the liquid, the more viscous the liquid is.

Experiment: Which liquid is the most viscous?

1 Use the following words to write an aim for this experiment.

liquid. find To viscous the most

Aim: _____

2 Fill in the missing vowels (a, e, i, o, u) to complete the list of equipment below.

Equipment

- 1 st__pw__tch
- 100-mL m__ __suring cyl__nd__r
- 1 m__rbl__
- L__qu__ds: shampoo, o__l, or__nge
- j__ __ce, glycerol, water

3 The steps below are out of order. Number the steps and then complete the prediction.

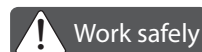
Method

- _____ Using a stopwatch, record how long it takes for the marble to reach the bottom.
- _____ **Gently** drop the marble from the top of the measuring cylinder into the first liquid.
- _____ Fill the measuring cylinder to the 100-mL line with the first liquid.
- _____ Record your results.
- _____ Pour the liquid back into its container and wash the measuring cylinder.
- _____ Repeat for the other four liquids.

Prediction

4 I predict that the marble will take the longest to travel through the _____.

Viscosity of liquids (2)



Read the Activity sheet 'Viscosity of liquids (1)', then complete the following questions.

1 Draw a scientific diagram of the experiment.

Scientific diagrams

When drawing a scientific diagram, remember to:

- use a ruler
- use a pencil
- draw a cross-section (2D)
- include labels.

2 Collect the equipment and complete the experiment. Record your results in the table below.

Liquid	Time taken to reach the bottom	Order of viscosity (1 = most viscous; 5 = least viscous)

3 In your workbook, draw a bar graph of your results. Use a different colour for each liquid and remember to leave a space between each bar.

4 Complete the conclusion.

Conclusion: The liquid with the highest viscosity was _____. It took _____ for the marble to travel through it. _____ has the lowest viscosity. It took _____ for the marble to travel through it.

What is diffusion?

Diffusion occurs when particles move from where there are lots of particles (high concentration) to where there are fewer particles (low concentration). Solids cannot diffuse because their particles are too close together and are strongly attracted to each other. Liquids and gases are able to diffuse. Gases diffuse the quickest because their particles are free to move quickly in different directions. When a liquid or gas is heated, it diffuses more quickly.

 **1** Read the following passage and circle the correct word in each pair.

The (movement/shrinking) of particles is called diffusion. A substance spreads out from an area where there is (more/less) of the substance to an area where there is (more/less). Gases diffuse (quickly/slowly) because the particles are moving quickly in all directions. Liquids can also diffuse but the particles move more (slowly/quickly).

Solid particles cannot diffuse because they cannot move, they can only (vibrate/slide over each other). An odour is a (gas/liquid) that is given off by a substance. You can smell perfume because the scent particles diffuse through the air to your nose in the form of a (solid/gas). If you drop a little bit of dye into a jar of water the colour will spread slowly through the water. This is an example of diffusion in (liquids/gases). When particles are (heated/cooled), they diffuse more quickly.

2 List three examples of diffusion.

3 Choose the correct ending for each of the following sentences.

Sentence start	Sentence endings		
Diffusion means	contracting.	spreading out.	expanding.
A substance diffuses from an area of high concentration to an area of	the same concentration.	high concentration.	low concentration.
Gases diffuse	quickly.	slowly.	only at very high temperatures.
A dye will diffuse more quickly in	cold water.	room-temperature water.	hot water.

Objectives

- To understand that when you heat matter, particles gain energy, which causes them to move more quickly and push apart
- To understand that when you cool matter the particles lose energy, which causes them to slow down and move closer together
- To identify examples of the different changes of state
- To identify the boiling and melting points of different substances and the effect of temperature on a substance's state of matter
- To interpret experimental results displayed in graphic form

Prior knowledge

Students should already be familiar with the differences between solids, liquids and gases in terms of properties and arrangement of particles. Students should also be aware that changes to materials such as melting, freezing and evaporation are reversible.

Background

When you heat matter, you give the particles energy. This causes them to move more quickly and push apart, eventually causing the substance to change state. The substance changes from a solid to a liquid (melting) and a liquid to a gas (evaporation). The type of material or substance does not alter; it just changes from one state to another. As the substance cools, the particles lose energy and slow down. The substance then changes from a gas to a liquid (condensation) and a liquid to a solid (solidification). Changes of state are reversible.

Starter activity

Place an ice block, a cup of water and a boiling kettle on the bench. Ask students to identify similarities (e.g. same substance) and differences (e.g. different states of matter) between them. Ask students how we can change water from one state to another (e.g. heat it up or cool it down), and have them suggest the names of some of the changes of state.

Activity sheets

The Activity sheet 'Heating up' introduces students to melting and evaporation. Have students label the

diagram showing the changes of state when matter is heated. Students can then list four everyday examples of substances changing state when heated, before identifying the change of state for the examples provided. True/false statements conclude the activity.

The Activity sheet 'Cooling down' explains the changes of state that occur when matter is cooled. Students label the flow chart and then separate out squashed sentences.

The Activity sheet 'Boiling and melting points' looks at what is meant by the boiling and melting point of a substance. Students identify the correctly spelt words in a passage and then answer questions about the melting and boiling points of four different substances.

The Activity sheets 'Temperature and state (1) and (2)' explore an experiment conducted to observe changes in temperature as a beaker of ice is heated up.

Students label the scientific diagram provided and order the steps involved in the experiment. Students view the results of the experiment in graphic form and are then asked to interpret this information.

Extension ideas

- Students could conduct an experiment on the boiling points of different liquids or the effect of salt on the boiling point of water.
- Students could perform an experiment to observe the effect of salt on the freezing point of water by making ice-cream in a zip-lock bag.

Selected answers

Boiling and melting points 2 a gas b liquid c below d 1535°C e -183° f boiling point; yes g oxygen

Temperature and state (2) 4 a solid b melting c heating up d liquid e boils f evaporation



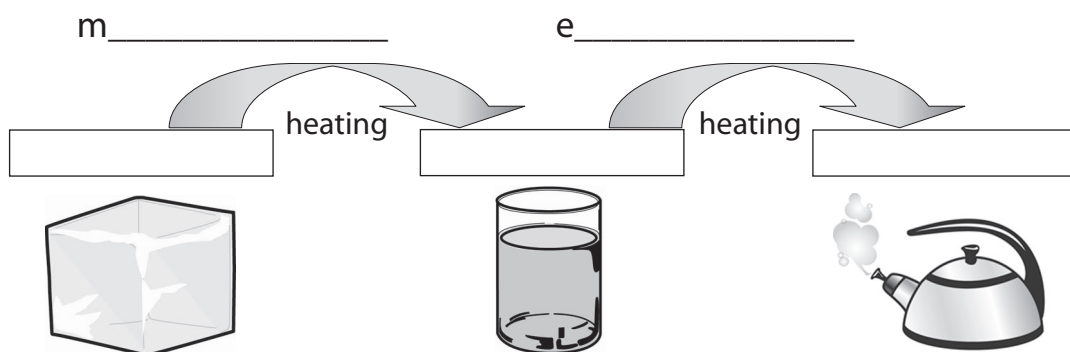
Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Heating up

When a substance changes from one state to another, the type of substance does not change. The particles in the substance do not change either. The only things that change are how quickly the particles can move and the amount of space between them. When you heat up a substance, the particles gain **energy**, causing them to move faster and spread out. This changes the substance from a solid to a liquid, or from a liquid to a gas.

-  **1** Use the words in the word bank to label the diagram below.



Word bank

evaporation
liquid
gas
melting
solid

- 2** List four everyday examples of substances changing state as they are heated.

- 3** Decide whether each of the actions listed in the table below is an example of melting or evaporation.

Action	Melting or evaporation
clothes in a dryer	
heating butter in a pan	
kettle boiling and producing steam	

- 4** Decide whether each of the following statements is true or false.

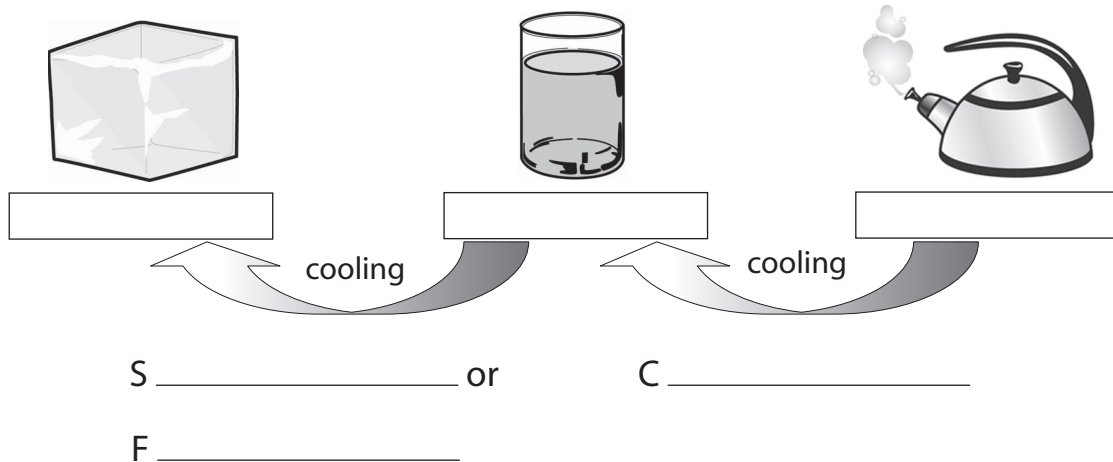
- a** When a substance is heated it gains energy. _____
- b** When a substance is heated the particles get bigger. _____
- c** When a substance is heated the particles move further apart. _____
- d** The change of state from a solid to a liquid is called melting. _____

Cooling down

As a substance cools, its particles lose **energy** and slow down. As they slow down, the particles become more **attracted** to each other, and move closer together. This changes the substance from a gas to a liquid, or from a liquid to a solid.



1 Use the words in the word bank to label the diagram below.



Word bank
condensation
liquid
gas
solidification
solid
freezing

2 Separate the words below to make sentences, then write out each sentence on the line below.

When a gas changes to a liquid it is called condensation.

When a liquid changes to a solid it is called solidification or freezing.

The change of state from a gas to a liquid is caused by cooling.

When a substance is cooled the particles lose energy.

When gas particles are heated they spread out and move quicker.

When liquid particles are cooled they move more slowly.

Boiling and melting points

 1 Read the following passage and circle the correctly spelt word in each pair.

The (boiling/boling) point of a substance is the (temperatur/temperature) at which a substance changes state from a (liquid/liqid) to a gas. The melting point of a substance is the temperature at which a substance changes state from a (soilid/solid) to a liquid. Different substances have different melting and boiling points. For (pure/prue) water, the melting point is 0°C and the boiling point is 100°C.

If the temperature is above a substance's boiling point, the substance is a (gas/gase). If the temperature is between a substance's melting and boiling point, the substance is a liquid. Below its melting point, a substance is a solid. Substances that have a (mealting/melting) and boiling point below room temperature (around 20°C) are usually gases.

2 Use the information in the following table to answer the questions below.

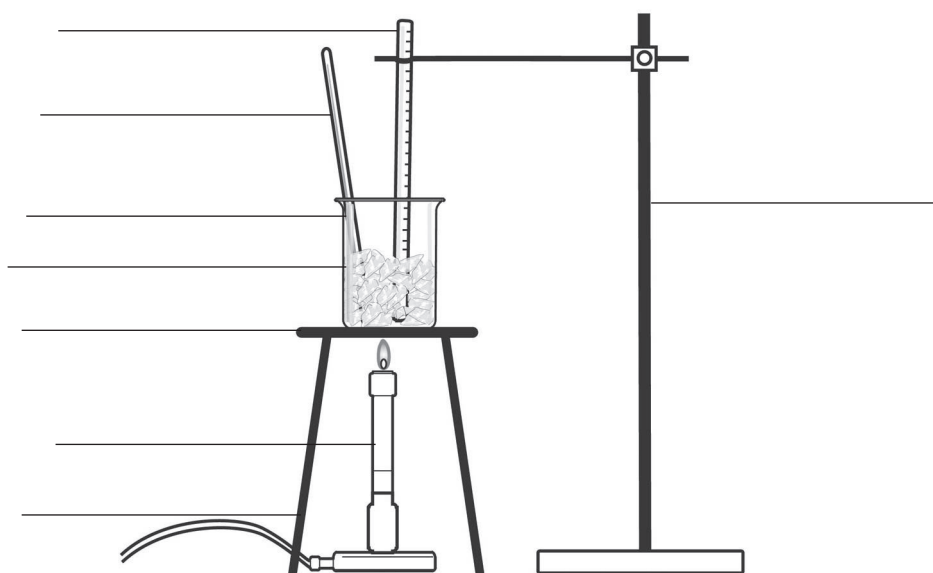
Substance	Melting point °C	Boiling point °C
Iron	1535	2750
Mercury	-39	357
Pure water	0	100
Oxygen	-218	-183

- What state of matter is oxygen at 25°C? _____
- At 25°C, what state of matter is mercury? _____
- Is 25°C above or below the melting point of iron? _____
- What temperature must iron be heated to before it will change from a gas to a liquid? _____
- What temperature does oxygen have to be before it will change from a gas to a liquid? _____
- Is 25°C above the melting point but below the b_____ p_____ for water? _____
- Which substance listed in the table has the lowest melting point? _____

Temperature and state (1)

Poppy conducted an experiment to observe the temperature changes of crushed ice as it is heated. The scientific diagram below shows how her experiment was set up.

-  1 Use the word bank to label the equipment used.



Word bank

beaker
tripod
retort stand and clamp
stirring rod
Bunsen burner
crushed ice
thermometer
gauze mat

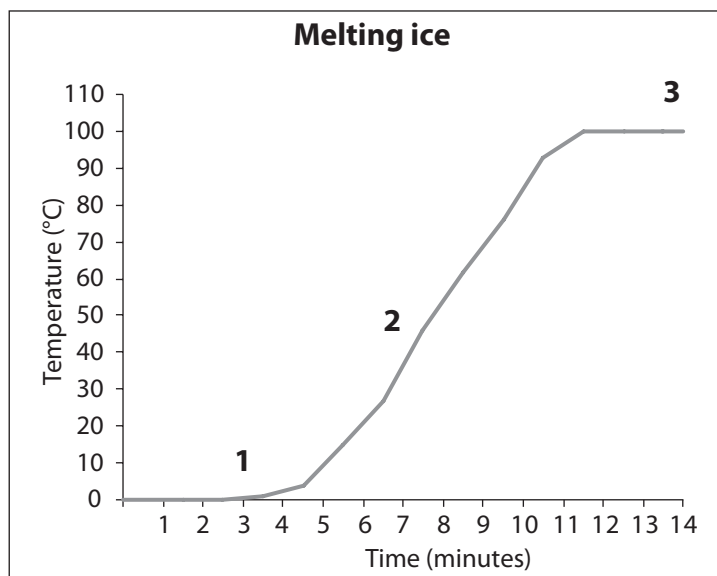
- 2 Below is a list of the steps that Poppy followed. Number the steps in the order in which they were completed.

- _____ Record the temperature every minute.
- _____ Place thermometer in ice without touching the bottom of the beaker.
- _____ Set up the tripod, gauze mat and Bunsen burner on a heatproof mat.
- _____ Use the stirring rod to constantly stir the ice slowly.
- _____ Record the temperature of the ice before lighting the Bunsen burner.
- _____ Place beaker on gauze mat.
- _____ Light the Bunsen burner and heat on a medium flame.
- _____ Half fill the beaker with crushed ice.
- _____ When the water begins to boil, continue heating for another 3 minutes.

- 3 Why do you think that Poppy made sure the thermometer didn't touch the bottom of the beaker? _____

Temperature and state (2)

Poppy used her results to plot a graph showing temperature against time. This is called a heating curve.



1 Use the words in the word bank to answer the questions below.

Word bank

boils solid liquid melting evaporation heating up

- What is the state of matter indicated at Stage 1? _____
 - What do you think is happening at Stage 1? _____
 - What do you think is happening at Stage 2? _____
 - Stage 2 represents what state of matter? _____
 - What happens to the liquid at 100°C? _____
 - What change of state is happening at Stage 3? _____
 - Circle the melting point and boiling point on the graph. Use blue for the melting point and red for the boiling point.
- 2 Write down one thing you learnt from Poppy's experiment.

Objectives

- To understand that an element is only made up of one type of atom
- To model the arrangement of particles in elements and mixtures of elements
- To recognise that elements can be represented by symbols
- To be able to locate elements on the periodic table

Prior knowledge

Students should be aware that all matter is made up of particles called atoms. They should also understand how particles are arranged in solids, liquids and gases.

Background

All matter is built from particles called atoms. A material in which all the atoms are the same is called an element. Elements have their own special properties. Elements are recorded on a chart called the periodic table. Metals make up most of the elements and are listed on the left-hand side of the periodic table. Non-metals are located on the right-hand side. We use chemical symbols to represent elements. Chemical symbols have one, two or three letters.

Starter activity

- Place a selection of common elements that students would recognise at the front of the classroom. Ask students if they know what the items have in common. Explain that they are all elements, and that each element is made up of one type of atom. Ask a student to place the elements into two different groups. They may divide them into groups based on their state of matter, or into metals and non-metals. Have students brainstorm how many metals and non-metals they can come up with.
- Listen to or view the Element Song to introduce the elements and the periodic table. For example: <http://www.youtube.com/watch?v=OduTDUGeAXE>

Resource and Activity Sheets

The Activity sheet 'Atoms and elements' explains the term 'element'. Students use the word bank to complete a passage and then unscramble important terms. They then consolidate the information by

identifying the difference between an element and a mixture of elements.

The Resource sheet 'The periodic table (1)' and the Activity sheet 'The periodic table (2)' develop students' understanding of the periodic table. Refer to the periodic table on the inside front cover when completing these work sheets. Students find the chemical symbols for different elements and determine if an element is a metal or non-metal. Students should recognise that if a substance is not on the periodic table it is not an element.

The 'Naming elements' Activity sheet presents a table containing clues to some of the elements in the periodic table. Students complete the table by identifying each element, then determining whether each is a metal or non-metal. Refer to the periodic table on the inside front cover when completing this work sheet.

In the Activity sheet 'Elements in the human body' students complete a crossword reinforcing the roles of different elements in the human body.

Extension ideas

- Students could research different elements and produce a poster about their chosen element.
- Students could conduct an experiment producing and testing for hydrogen or oxygen gas.
- Students could play 'Element Bingo'.

Selected answers

Naming elements Symbol column: He, Li, Br, Ba, Po, Hg, Ar, Sn, Au, Fe, B, Es, Eu, Cu

Elements in the human body Across: 3 oxygen 5 hydrogen 6 calcium 7 phosphorus 8 fluorine 10 iodine 11 zinc **Down:** 1 potassium 2 sodium 4 nitrogen 6 carbon 9 iron



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

Atoms and elements

 1 Use the words in the word bank to complete the passage below.

Everything around us, and all the things we use are made of m_____. All materials are made up of tiny p_____ called **atoms**. Although there are millions of different materials that exist, they are made up of just over 100 different elements. An element is a substance that cannot be split into anything simpler. E_____ contain only o_____ type of a_____. For example, iron is an element made up of iron atoms. Carbon, which is found in all l_____ things, is made up of c_____ atoms. Marble is not an element, because the atoms in marble are not all the s_____.

Word bank

living
particles
atom
Elements
one
carbon
materials
same

2 Unscramble these important words used in the passage above.

mstao _____

mtseenel _____

tmlareasi _____

rsceaptil _____

3 Circle the correct answer.

Instead of saying particle, sometimes scientists use the word element / material / atom / carbon.

4 Look at the diagrams below and decide which is an element and which is a mixture of elements. Label them.

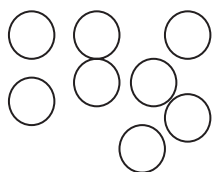


Diagram 1

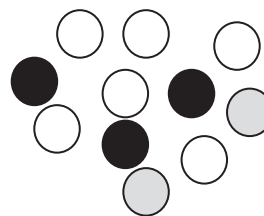


Diagram 2

5 Diagram 1 is made up of _____ type of element.

6 Diagram 2 is made up of _____ different elements.

The periodic table (1)

Elements such as carbon, gold and copper have been known since ancient times. Hydrogen was discovered in 1766. By 1800, scientists had started sorting the elements into groups. At that time many elements had not yet been discovered. In 1869, a Russian scientist named Dimitri Mendeleev ordered the elements according to how heavy they were. He displayed this information in a chart and called it the **periodic table**.

Look at the periodic table on the inside front cover of this book. The periodic table contains more than 100 elements, but only 92 of them are found naturally on earth. The rest are man-made. Elements that have similar properties are grouped together in columns in the table. For example, helium (He) has similar properties to neon (Ne) – both are gases that do not react – so they are grouped together.

The elements grouped on the left-hand side of the periodic table are **metals**.

The **non-metals** are grouped on the right-hand side.

Each element has a **chemical symbol**. Some symbols have one letter, while others have two or three. For example, C is the symbol for carbon. Chlorine, which was discovered after carbon, has the symbol Cl. Some symbols come from the element's Latin name. The symbol for iron is Fe, which comes from its Latin name, *Ferrum*.

The periodic table (2)

 Use the Resource sheet 'The periodic table (1)' and the periodic table on the inside front cover to answer the following questions.

1 What is listed in the periodic table? _____

2 Who first developed the periodic table? _____

3 In what year was the periodic table first developed? _____

4 Find out the symbol for the following elements. If the symbol comes from the Latin name, write the symbol in red pen.

a magnesium _____ e sodium _____ i carbon _____

b oxygen _____ f aluminium _____ j gold _____

c nitrogen _____ g iodine _____ k potassium _____

d sulphur _____ h zinc _____ l chlorine _____

5 Which of the elements listed in Question 4 are non-metals?

6 Sort the substances below into the correct groups.

water iron sand air carbon aluminium
silver oxygen carbon dioxide helium plastic
sulphur calcium steel sodium salt argon

Element	Not an element

Naming elements

 Use the Resource sheet 'The periodic table (1)', the periodic table on the inside front cover, and the word bank below to help you solve the riddles and complete the table below. Each riddle gives a clue about what the name of the element sounds like.

For each riddle:

1 Write down the element's full name and its symbol.

2 Classify it as a metal or non-metal.

Word bank

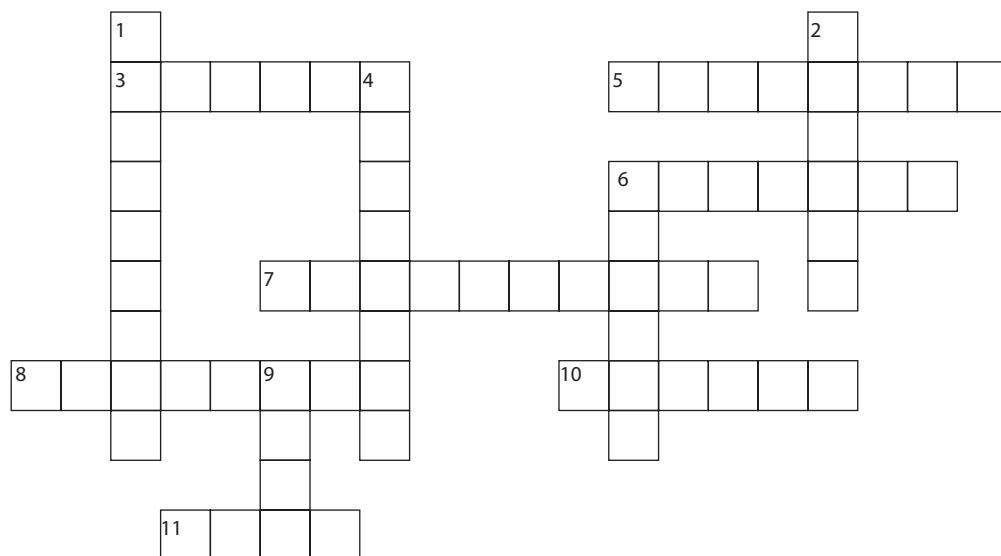
Po	He	Li	Es	B	Au	Br
Hg	Cu	Sn	Fe	Eu	Ar	Ba

Riddle	Element name	Symbol	Metal or non-metal
What you hope a doctor can do			
What a weightlifter does			
Male sibling of someone			
What a gravedigger has to do			
A kind of sausage, also known as devon			
Closest planet to the sun			
Once someone leaves they ...			
Rhymes with what you put rubbish in			
What you get when you win			
Used to remove wrinkles from clothes			
Dull and unexciting			
A very clever element			
What you do with a lasso			
A slang name for a policeman			

Elements in the human body



1 Use the words in the word bank to complete the crossword puzzle.



Word bank

sodium
carbon
oxygen
iron
potassium
calcium
hydrogen
fluorine
nitrogen
zinc
iodine
phosphorus

Across

- 3 O – needed for breathing
5 H – element that makes up water
6 Ca – needed for healthy bones
7 P – needed for muscle growth and nerve function
8 F – often added to water supply to improve health of teeth
10 I – needed for producing hormones
11 Zn – helps with colds

Down

- 1 K – important for muscle contractions
2 Na – found in sweat
4 N – gas found in your lungs
6 C – element found in all living things
9 Fe – mostly found in the blood

2 Match the sentence halves below to complete the following statements.

Potassium is an important

are two gases found in your lungs.

Carbon is found

your blood is iron.

An important mineral stored in

in all living things.

Oxygen and nitrogen

mineral needed for muscle contractions.

Objectives

- To locate different elements on the periodic table and identify them as metals or non-metals
- To explain the properties of metals and non-metals
- To develop students' ability to make observations and record and analyse data

Prior knowledge

Students need to have a basic understanding of the periodic table.

Background

Elements are grouped on the periodic table. Metals are found on the left-hand side and non-metals are found on the right-hand side. Most metals are strong, shiny, can be beaten into shape and bend without snapping. Metals are also good conductors of heat and electricity. Non-metals are considered the opposite of metals. They are usually light, are not shiny, are brittle and cannot be beaten into shape. Most non-metals do not conduct electricity or heat. Only about one-sixth of elements are non-metals.

Starter activity

- Play a game of 'What am I?'. Describe the characteristics and uses of common metals and non-metals.
- Provide students with an envelope containing squares from a pictorial periodic table depicting common metals and non-metals. There are many pictorial periodic tables on the internet. Have students divide the squares into metals and non-metals and use the pictures to brainstorm uses of metals and non-metals, or the characteristics of metals and non-metals.

Activity sheets

In the Activity sheet 'Metals and non-metals' students classify different elements as either metals or non-metals. Students are then given a group of elements and need to identify the odd one out. Refer to the periodic table on the inside front cover when completing this work sheet.

'Properties of metals' explores the characteristics of metals. Students complete a cloze exercise, then

re-read the completed passage to identify the main properties of metals and record these in the table provided. They then match a set of phrases to consolidate their learning.

'Properties of non-metals' introduces the idea that non-metals have the opposite characteristics of metals. Students summarise notes about the characteristics of non-metals and then unjumble sentence parts describing the uses of some common non-metals.

The Activity sheet 'Carbon' explains the different forms in which carbon can be found. Students complete a cloze exercise by filling in the missing vowels. The lesson concludes by asking students to predict which substances from the list provided contain carbon.

'Reactive metals' requires students to make observations and record and analyse data while you conduct a demonstration of how different metals react in water. Students identify the equipment used, order the steps in the method and identify risks associated with the demonstration. They then conclude which metal was the most reactive. Equipment required includes metal tongs, 1L beaker, safety glasses, scalpel, board and pieces of sodium, potassium and lithium. Ensure all students wear safety glasses, and do not use pieces of metal larger than the size of a pea.

Extension ideas

- Students could make a metal collection by gathering everyday items. They could attach tags explaining what metal each item is made out of and the reasons why it is made from that metal.
- Students could play 'Snap' using cards displaying properties of metals and non-metals and definitions of these properties, along with the names of metals and non-metals and their uses in words or pictures. Students identify matching elements and properties in order to play 'Snap'. The same cards can also be used to play 'Memory'.

Selected answers

Carbon 3 Yes, Yes, No, Yes, No, No



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Metals and non-metals

 Use the periodic table on the inside front cover to complete the questions below.

1 Group the elements in the word bank into metals and non-metals.

Metals	Non-metals

Word bank

manganese

S

chlorine

Cu

Al

iodine

Na

argon

carbon

Zn

boron

K

Iron

O

helium

silicon

Ag

Cr

2 Circle the element that is the odd one out in each group. Give a reason for your answer.

a magnesium, silver, bromine, nickel

b Ca, Ba, Ni, Mg

c Cr, Cu, Zn, Ag

d oxygen, helium, nitrogen, sodium

3 Name three metals that have similar properties to magnesium.

Properties of metals

Substances are often classified into groups containing similar substances. Metals are one such group.

 **1** Use the words in the word bank to complete the passage below.

Metals are grouped on the l_____ -hand side of the p_____ table. They are good conductors of h_____ and e_____. Metals are usually h _____ and strong but they are also f_____, which allows them to be shaped. Metals are s_____ when polished, but some can tarnish. They also have h_____ melting and b_____ points. M_____ is the only metal that is not a s_____ at room temperature. Metals are heavy for their size. We say that they are d_____.

Word bank

shiny
Mercury
dense
left
heat
high
solid
electricity
periodic
flexible
hard
boiling

2 Use the information above to complete the table below.

Properties of metals
•
•
•
•
•
•
•
•

3 Match the phrases below.

Metals are good

melting and boiling points.

Metals are flexible

that is not a solid at room temperature.

Mercury is the only metal

and can be shaped.

Metals have high

conductors of heat and electricity.

Properties of non-metals

Non-metals have the opposite characteristics to metals. For example, metals are shiny but non-metals are dull. Non-metals:

- do not conduct heat and electricity very well
- are brittle (this means they may shatter if you hit them)
- are usually dull
- have low melting and boiling points
- are usually gases at room temperature
- are light for their size
- are softer and weaker than metals
- can be solids, but only bromine is a liquid at room temperature.

 **1** Use the notes above to write a paragraph in your workbook describing the properties of non-metals.

2 Unjumble the sentence parts and then rewrite each sentence in the correct order.

a to make Sulphur is used matches, fungicides and sulfuric acid.

b for bleach Chlorine is used and disinfectant.

c fertiliser used in Nitrogen is snap-freezing vegetables. and for

d prevent tooth decay. in toothpaste to Fluorine is used

e lighter than air, is used to fill balloons so when it Helium is they float.

Carbon

Carbon is a common **non-metal** that can be found in a number of different forms, including diamonds, graphite and charcoal. Different forms of carbon have different properties.



 **1** Fill in the missing vowels (a, e, i, o, u) to complete the following passages.

D _ _ m _ nd is the hardest known substance on earth. It is used in dr _ ll bits and s _ w blades because it is so h _ rd. It is also used in j _ w _ ll _ ry because of its sp _ rkl _.

Gr _ ph _ t _ is the most common form of c _ rb _ n. It is a soft bl _ ck solid. Graphite is a sp _ c _ _ l n _ n-m _ t _ l because it conducts el _ ctr _ city. It is used in spark-pl _ g leads and in b _ tter _ es.

Ch _ rc _ al is a blackish substance that forms when w _ _ d and other pl _ nt and _ nim _ l matter is b _ rnt. It is used as a fuel s _ _ rce, most commonly in c _ _ king. It can also be used in f _ lt _ rs. Charcoal t _ blets are used to tr _ _ t st _ m _ ch problems.

2 Draw a line to match each form of carbon to its uses.

Form of carbon

graphite

diamond

charcoal

Use

fuel source and filters

drill bits and saw blades

batteries and spark-plug leads

3 Rosie conducted an experiment to find out which items contain carbon. Using metal tongs, she burnt different substances over a Bunsen burner. If the substance burnt and turned black, Rosie concluded that it contained carbon. Predict which of the substances below contain carbon. (Hint: Think about where each substance came from.)

Substance	woollen cloth	sugar cube	aluminium foil	bread	steel wool	glass
Contains carbon						

Reactive metals



Some metals are more **reactive** than others. The most reactive metals are listed on the left-hand side of the periodic table. Your **teacher** is going to demonstrate how three different metals react in water.

Experiment: How do different metals react in water?

1 Fill in the missing vowels (a, e, i, o, u) to complete the list of equipment.

- m _ t _ l t _ ngs
- 1L b _ _ k _ r
- s _ f _ ty gl _ ss _ s
- sc _ lp _ l
- b _ _ rd
- pieces of s _ d _ um, pot _ si _ m and l _ thium

2 The steps in the method below are out of order. Number the steps in the order they should be done.

- _____ Using the scalpel, cut a small piece of lithium on the board (no larger than the size of a pea).
- _____ Repeat for sodium and potassium.
- _____ Put on safety glasses and half-fill the beaker with water.
- _____ Record observations in table.
- _____ Using metal tongs, drop lithium into the water.

3 Identify one risk with this experiment.

4 Record your observations in the table.

Observation	Lithium	Sodium	Potassium
colour			
floats or sinks in water			
reaction in water			

5 **Conclusion:** The least reactive metal was _____. The most reactive metal was _____.

Objectives

- To be able to model the arrangement of particles in elements, in molecules of elements and in compounds
- To recognise that elements and simple compounds can be represented by symbols and formulas
- To identify the elements in different simple compounds
- To draw molecular diagrams to represent simple compounds

Prior knowledge

Students should know the symbols for common elements or be able to locate them on the periodic table.

Background

The millions of materials in the world are made up of just a small number of different elements joined together in different ways. Elements combine to form compounds. A compound is a material made up of more than one element. When atoms join up they form a molecule. A molecule can be made up of atoms of a single element – for example, an oxygen molecule has two oxygen atoms chemically joined together – or it can be made up of different atoms (a molecule of a compound). We can use symbols and formulas to represent compounds.

Starter activity

- Show students a model of a water molecule. Explain that this represents one of the most common and important substances on earth. Ask students to guess what kind of molecule it is. Once they have identified that it is water, explain that the different-coloured balls in the model represent two different elements. Ask students to guess what they are. Also ask if there is another way of saying water (H_2O). Write this on the board and relate it back to the model of the molecule of water.
- Show the class an animated video that teaches students about the different elements of the periodic table, for example:
www.youtube.com/watch?v=d0zI0N8xjbM

Resource and Activity sheets

The Resource sheet 'Joining elements (1)' introduces students to chemical formulas and demonstrates how to draw molecular diagrams. Students use this information to assist them when completing the Activity sheet 'Joining elements (2)'. Students define key terms with the help of a word bank and complete a true/false activity. Students then identify elements and compounds to consolidate their understanding.

The Activity sheet 'Common compounds' provides students with information about two common compounds: water and table salt. Students order sentences to form a paragraph about water. They then need to join up the sentence parts about the compound sodium chloride.

The Activity sheet 'Drawing and making molecules' provides students with the opportunity to make different molecules of elements and compounds using a molecular kit. Students can make a number of different molecules and then complete the table provided. They finish off the lesson by identifying the type and number of atoms that make up different molecules. Students will need access to molecular kits to build their molecules.

The Activity sheet 'Writing chemical formulas' provides students with a basic introduction into how to write chemical formulas. They fill in missing vowels and then use the key and molecular diagrams to write formulas for different molecules.

Extension ideas

- Students could conduct an experiment to produce and test for carbon dioxide gas.
- Students could play 'Compound Bingo' using common compounds.

Selected answers

Writing chemical formulas 2 H_2O , SO_2 , $MgCl_2$, CO_2 , H_2S , CH_4



Work safely

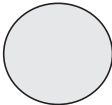
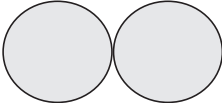
Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

Joining elements (1)

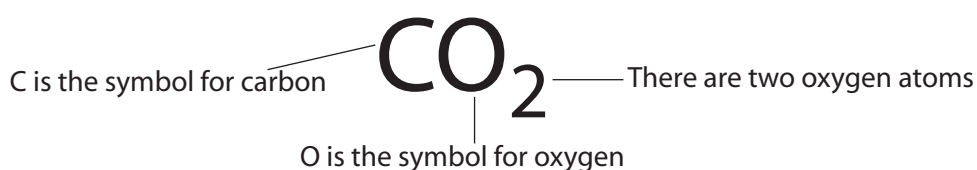
In some **elements**, such as oxygen, the atoms pair up. A particle that is made up of more than one atom is called a **molecule**. We can show this using the chemical symbols for the atoms involved. For example, oxygen molecules can be represented as O_2 . The number in the **formula** shows how many atoms of the type there are in the molecule.

We can represent molecules by drawing coloured circles to show the atoms. When we draw a molecule we draw the atoms of that type touching, to show that they are chemically joined together.

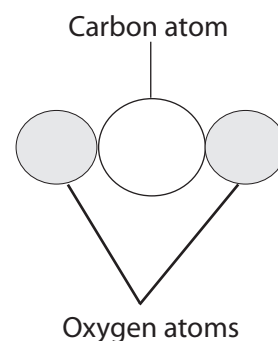
Element	Formula for molecule	Picture of molecule
oxygen – a non-metal gas 	O_2	 oxygen gas
nitrogen – a non-metal gas 	N_2	 nitrogen gas

When two or more atoms from different elements join together, they form a **compound**. The formula for a compound shows the types of elements that are in the compound and the ratio (how many atoms of each element there are).

Carbon dioxide is a compound made up of carbon and oxygen. It has the formula:



This formula tells us that a molecule of carbon dioxide is always made up of one carbon atom and two oxygen atoms. We don't need to add a one after carbon. A chemical formula only includes a number after an element symbol if there is more than one atom of that element.



Joining elements (2)

Read the Resource sheet 'Joining elements (1)', then answer the following questions.

 1 Use the words in the word bank to complete the table.

Word	Description
atom	Tiny p _____ that cannot be b _____ down into s _____ substances
element	Made up of only o _____ type of a _____
compound	Made up of t _____ or more e _____ joined together
molecule	Two or more a _____ joined together. If the atoms are the s _____ it is a m _____ of an element. If the atoms are d _____, it is a molecule of a c _____.

Word bank

atoms
broken
compound
two
particle
simpler
different
same
one
elements
atom
molecule

2 Decide whether each of the following statements is true or false.

- a An element is made up of only one type of atom. _____
 - b A compound can be made of atoms that are the same or different. _____
 - c A molecule is always made up of two or more different types of atoms. _____
 - d We use a formula to show the type of elements and the number of atoms of each element in a molecule. _____
 - e Nitrogen dioxide has the formula NO_2 . This tells us that there are two elements present, nitrogen and oxygen, and that one molecule contains one nitrogen atom and two oxygen atoms. _____
- 3 Circle the element symbols and underline the formulas of compounds in the box below.

SO_2 Cu Zn NaCl Cl H_2O K

Common compounds

Millions of different substances exist on earth. These substances are all made up of just 100 or so elements, which join up in different ways to make different **compounds**.



- 1** Water is one of the most common compounds on earth.
Number the following sentences about water in an order that makes sense, then rewrite the paragraph in your workbook.

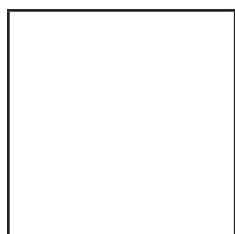


- _____ This means that each molecule of water has two atoms of hydrogen combined with one atom of oxygen.
- _____ Approximately 70% of an adult's body is made up of water.
- _____ Water is one of the most common compounds on earth.
- _____ Water is a common compound made up of the elements hydrogen and oxygen.
- _____ It is essential for all life on earth.
- _____ Scientists write the formula H_2O to represent the water molecule.

- 2** Draw lines to join up the sentence parts about table salt.

The scientific name for table salt	a compound.
Sodium chloride is	to flavour food.
A molecule of sodium chloride is made up	is sodium chloride.
The formula for table salt is	sodium, which is a metal, and chlorine, which is a non-metal.
Table salt contains two elements:	of one atom of sodium and one atom of chlorine.
Salt is used	NaCl.

- 3** Draw a diagram of a molecule of water and a molecule of table salt (sodium chloride).



Drawing and making molecules

Molecules are particles that are made up of more than one atom. In a molecule of an **element**, the atoms are the same. In a molecule of a **compound**, the atoms are different.

-  **1** Using a molecular kit and the periodic table, build the common molecules listed in the table below, then complete the table. When drawing the molecules, use the same colour as the atoms in the molecular kit.

Molecule name	Formula	Molecule diagram	Element or compound
oxygen	O ₂		Element
carbon dioxide	CO ₂		
ozone	O ₃		
water	H ₂ O		
table salt	NaCl		

- 2** Next to each molecular compound, write down the name and number of each type of atom that is joined to make the molecule. The first one has been done for you.

- | | |
|--|------------------------------------|
| a water (H ₂ O) | 2 hydrogen atoms and 1 oxygen atom |
| b sulfur dioxide (SO ₂) | _____ |
| c carbon monoxide (CO) | _____ |
| d hydrochloric acid (HCl) | _____ |
| e table salt (NaCl) | _____ |

Writing chemical formulas

Writing chemical formulas can be tricky, but it is a lot easier if you remember a few rules.

 **1** Fill in the missing vowels (a, e, i, o, u) to complete the rules for writing formulas.

Rule 1: When an el ___ m ___ nt symbol has one l ___ tt ___ r, use a c ___ p ___ tal letter.

Rule 2: When an element symb ___ l is made up of tw ___ letters, use a capital l ___ tter and a l ___ wer case letter.

Rule 3: When there is m ___ r ___ than one at ___ m of an element, you must show this by writing a sm ___ ll number after the ___ lem ___ nt symb ___ l.

Rule 4: The m ___ tal is usually written f ___ rst.

Rule 5: Some compound names give you cl ___ s about the ___ rd ___ r of the ___ l ___ ment symb ___ ls.

					
Hydrogen	Oxygen	Carbon	Sulphur	Chlorine	Magnesium
H	O	C	S	Cl	Mg

KEY FOR ATOMS

2 Use the key above to write the formulas for the molecules below.



water _____



sulfur dioxide _____



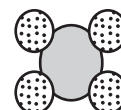
magnesium
chloride _____



carbon dioxide _____



hydrogen sulfide _____



methane _____

Objectives

- To recognise the differences between pure substances and mixtures and identify examples of each
- To explain the difference between soluble and insoluble substances
- To identify the solvent and solute in solutions
- To develop students' ability to plan and safely conduct experiments and to record and analyse data
- To identify factors that affect solubility

Prior knowledge

Students should understand that some solids can dissolve in different liquids and others can't. They should also understand that dissolving is not the same as melting. Students will need to be able to draw a line graph and a scientific diagram.

Background

Pure substances can be elements or compounds. Mixtures are two or more different substances that have been mixed together but not chemically combined. In pure substances the elements are present in fixed amounts, whereas in mixtures there is no set ratio. A solution is an example of a mixture that forms when a soluble substance (solute) dissolves in another substance (solvent). The solubility of a substance can be affected by a number of different factors, including temperature and the size of the solute particles. If you keep adding more solute to a solution, eventually no more will dissolve, and some solid will be left in the bottom. When this happens, the solution has become saturated.

Starter activity

Demonstrate making a hot chocolate. Ask students:

- to identify the different components of the drink
- which components are made up of only one substance and which are made up of more than one substance. Then explain the difference between a pure substance (sugar) and a mixture (hot chocolate powder, milk).
- to identify the substances that dissolve (sugar, hot chocolate powder) and the substance that dissolves them (hot water). Introduce the terms 'solute' and 'solvent'.

Activity sheets

The Activity sheet 'Mixtures and pure substances' explains the differences between a pure substance and a mixture. Students identify spelling errors in a passage before completing a matching phrases activity. Students finish the lesson by classifying a number of substances as either pure substances or mixtures.

In the 'Solutions' Activity sheet students complete a crossword puzzle to familiarise themselves with key words associated with solutions.

The Activity sheets 'Investigating solvents (1) and (2)' develop students' skills in safely conducting experiments and recording and analysing data. Students investigate which common substances are soluble in water and alcohol. Substances tested are salt, crushed chalk, jelly crystals, sugar, ballpoint-pen ink and iodine. Ensure students complete questions 1 and 2 about risks before conducting the experiment. Close supervision is required to ensure student safety.

The Activity sheet 'Factors affecting solubility' introduces students to different factors that affect solubility. Students complete a cloze activity to explore the topic. They then examine a table of solubilities of different solutes, then use this information to answer some questions.

Extension ideas

- Students could conduct an experiment to investigate the differences between soluble and insoluble substances.
- Students could conduct an investigation, testing which factors affect the rate at which sugar dissolves.

Selected answers

Solutions Across: 2 paint 3 solution 5 solute 6 salt 7 soluble **Down:** 1 dissolves 3 sugar 4 insoluble 5 solvent



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Mixtures and pure substances

 **1** Rewrite the words in the passage that have been spelt incorrectly.

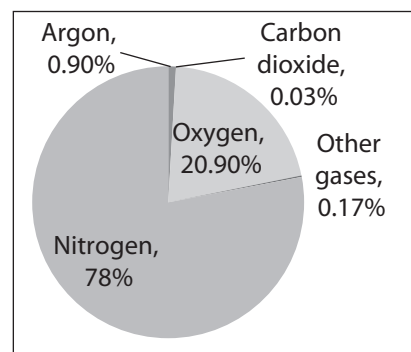
A pur () substance is a substance that has not been mixed with anything else.

Elements can be pure substances (). For example, if you had a jar of oxygen that was not mixed with anything else, you would have pure oxygen.

Compounds can also be pure. Watter () is pure if it doesn't contain anything except water. **Mixtures** contain two or more substances that are mixed up, but not chemicaly () joined.

A **substance** is a mixture if it can be separated into other substances. Mixtures can be made up of both eliments () and compounds. Air is a mixture () that contains the elements nitrogen, oxigen () and argon. It also contains the compunds () water and cabon () dioxide.

Unlike for compounds, you cannot write a formula for a mixture.



Air is a mixture of different substances.

2 Draw lines to match the sentence halves below.

- | | |
|---------------------------------|--|
| A pure substance | both elements and compounds. |
| Both elements and compounds can | two or more substances not chemically joined together. |
| A mixture is made up of | is not mixed with anything else. |
| Air is a mixture made up of | be pure substances. |

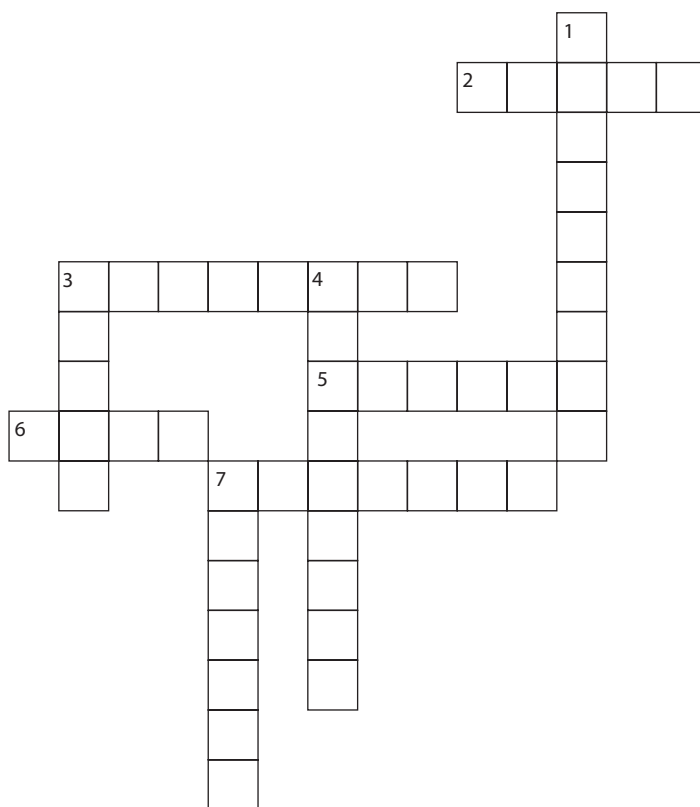
3 Which of the following substances are pure substances? Circle them, and then underline the mixtures.

gold tap water salt cola
oxygen sugar coffee

Solutions

- If a solid dissolves in water it is **soluble**. If it doesn't dissolve, it is **insoluble**.
- The substance that dissolves another substance is called the **solvent**.
- When one substance dissolves in another, a **solution** forms.
- Liquids and gases can also dissolve in a solvent. In soft drinks, carbon dioxide dissolves in water, forming bubbles.

 **1** Use the words in the word bank to complete the crossword puzzle.



Word bank

salt
solution
solvent
paint
dissolves
insoluble
sugar
solute
soluble

Across

- 2** Mineral turpentine is a solvent often used to dissolve _____.
- 3** Forms when one substance dissolves in another
- 5** A _____ dissolves in another substance.
- 6** In seawater, the solute is the _____.
- 7** If a substance dissolves it is _____.

Down

- 1** Instant coffee _____ in hot water.
- 3** A soluble substance used when making tea and coffee
- 4** Sand is _____ in water.
- 7** The substance that dissolves another substance is called the _____.

Investigating solvents (1)



A **solvent** is a substance, often a liquid, that is used to dissolve another substance to form a solution. Alcohol and water are two common solvents.

Experiment: Which substances are soluble in water and alcohol?

Equipment

- 7 test tubes
- safety glasses
- marker pen
- test-tube rack
- spatula
- rubber stopper
- stopwatch
- 6 different substances: salt, crushed chalk, jelly crystals, sugar, ballpoint-pen ink and iodine
- spotting tray
- dropper bottle of ethanol (form of alcohol) **NOTE: POISON, do not drink.**
- piece of paper

1 Use the words in the word bank to complete the method.

Method

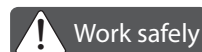
- Put on _____ glasses and, using the spatula, collect a sample of each of the _____ and put in spotting tray.
- Using the _____ pen, label each test tube with the name of the substance.
- Place the first substance in the _____ and half fill with water.
- Place rubber _____ on end of test tube and shake for 30 seconds.
- Record whether the substance is _____, insoluble or slightly soluble (if some _____) in water.
- Repeat for other substances. For ballpoint-pen ink, draw a dot on a piece of _____ and place four drops of water on paper.
- Repeat tests with ethanol.

Word bank

stopper
paper
safety
marker
soluble
substances
test tube
dissolves

NOTE: Do not dispose of iodine down the sink. Leave for your teacher to dispose of.

Investigating solvents (2)



Read the Activity sheet 'Investigating solvents (1)', then answer the following questions.

- 1 Identify two risks with this experiment. _____
- 2 State how you will address these risks. _____
- 3 In your workbook, draw a scientific diagram of the experiment. Use a ruler and pencil. Draw a cross-section, and remember to include labels.
- 4 Complete the experiment and record your results in the table below.

Substance	Soluble, insoluble or slightly soluble in water	Soluble, insoluble or slightly soluble in alcohol
salt		
crushed chalk		
jelly crystals		
sugar		
ballpoint-pen ink		
iodine		

- 5 Which substance/s dissolved the most easily in water? _____
- 6 Which substances were soluble in water but not in alcohol? _____
- 7 Which substances dissolved in alcohol but not in water? _____
- 8 Were there any substances that did not dissolve in either water or alcohol?

- 9 Write a simple conclusion about what you learnt from this experiment.

Factors affecting solubility

Solubility tells us how much of something can dissolve in something else. There is a limit to how much **solute** will dissolve in a **solvent**. When no more solute will dissolve, we say that the solution has become **saturated**.

 **1** Use the words in the word bank to complete the following statements.

Solubility is affected by the following factors:

- whether or not the solution is _____, and how quickly
- the type of _____ the solute is dissolved in
- the _____ of the solute particles. Smaller-sized particles dissolve more _____
- the _____ of the solvent.

Word bank

size
solvent
quickly
temperature
stirred

Different substances have different solubilities, and solutes become more soluble at higher temperatures.

Temperature	Potassium chloride Solubility g/100g water	Copper sulfate Solubility g/100g water	Sodium chloride Solubility g/100g water
10°C	31	19	34.5
30°C	37	24	36
50°C	42	33	37
70°C	49	48	38
90°C	53	65	39.5

- Which substance has the highest solubility at 50°C? _____
- Which substance's solubility is the least affected by changes in temperature? _____
- What is the solubility of potassium chloride at 30°C? _____
- Which substance has a solubility of 48g/100g water at 70°C? _____

Objectives

- To investigate a range of physical separation techniques such as filtration, evaporation, crystallisation and distillation
- To identify where filtration is used in and around the home

Prior knowledge

Students should have a basic knowledge of some solids that are soluble in water and some substances that are insoluble. Students need to understand that when a solid dissolves, even though we can no longer see it, it is still there.

Background

Substances in a mixture can be separated because of their different properties. A number of different separation methods include evaporation, crystallisation, distillation and filtration.

Starter activity

Present students with a scenario involving substances that have been mixed and ask them to consider how these substances might be separated. For example: 'You decide to mix three substances that you find in your dad's shed: beach sand, iron filings and marbles. When your father discovers what you have done, he is not happy and wants you to separate the substances. How can this be done?'

Resource and Activity sheets

The Activity sheet 'Filtration' introduces the separation method of filtration. A scientific diagram of filtration is provided for students to label. Students then complete true/false questions to consolidate their understanding before finishing off the lesson by identifying which of the substances listed can be separated from water using filtration.

The Activity sheet 'Filters in and around the home' looks at a variety of different filters that can be found around the home. Students match sentence halves that describe different filters used in and around the home and then identify the residue and filtrate for different filters.

'Evaporation and crystallisation' describes the differences firstly between filtration and evaporation and then between evaporation and crystallisation. Students start by choosing the correct word from the pairs, and then complete some true/false questions.

The Resource sheet 'Distillation (1)' outlines the process of distillation. Students then use this information to help them complete the Activity sheet 'Distillation (2)', which presents comprehension questions and asks students to order the steps of an experiment. A word bank is provided to help students label a scientific diagram of the experiment.

Extension ideas

- Students could be given a challenge similar to the starter activity. Give students a mixture of three substances (an insoluble substance, soluble substance and a liquid), then have students conduct an experiment to separate all three substances.
- Students could perform an experiment where they use distillation to make perfume.

Selected answers

Filters around the home: 2 tea leaves, liquid tea; dust from air, clean air; mineral deposits and impurities water; ground coffee, liquid coffee; dirt and impurities in oil, clean oil; fluff and hair, clean filtered air



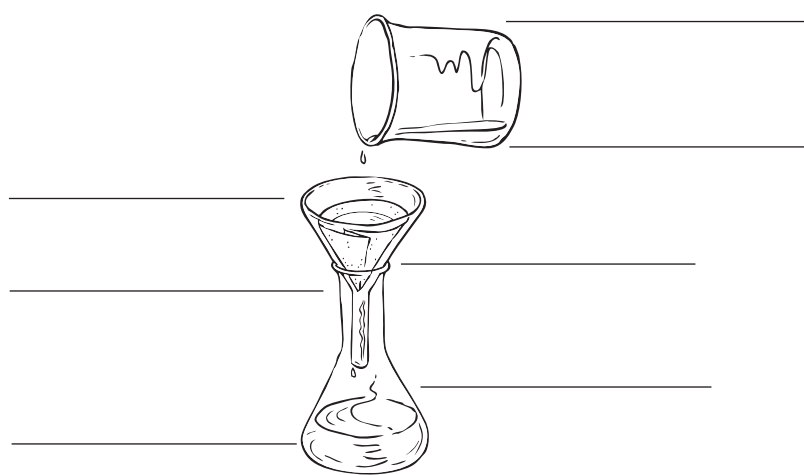
Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Filtration

Filtration is used to separate a mixture of an insoluble solid and a liquid. The mixture is poured through filter paper. Only the liquid passes through – this is called the **filtrate**. The solid that is collected on the filter paper is called the **residue**.

-  **1** Read the passage above, then use the words in the word bank to label the scientific drawing of filtration.



Word bank

mixture
filter funnel
residue
beaker
filter paper
conical flask
filtrate

- 2** Decide whether each of the following statements is true or false.

- a** Salt and water can be separated using filtration. _____
- b** The liquid that has been filtered is called the filtrate. _____
- c** Soluble solids can be separated from liquids using filtration. _____
- d** The solids that are collected in the filter are called the residue. _____
- e** Both the liquid and insoluble solid pass through into the filtrate. _____

- 3** Circle the substances that can be separated from water using filtration.

salt jelly crystals pepper

sugar baking soda cordial

soil

instant coffee tea leaves

Filters in and around the home

Many different types of filters are used around the home. Some filters remove solids from liquids and gases, and others, such as gas masks, remove gases from the air.

 1 Match the sentence halves below describing filters used around the home.

Gas masks are used to	to keep the ground coffee out of the drink.
Tea bags are an example of a filter – the water filters through the bag,	to remove large mineral deposits and impurities from water.
Air conditioners have filters	collect fluff and hair from clothes.
Water jugs use a charcoal filter	remove poisonous chemicals and bacteria from the air.
Coffee machines use a filter	that remove dust from the air.
Oil filters remove	harmful substances that could damage your car.
Clothes dryers have lint filters, which	releasing the tea but keeping the ground tea leaves from ending up in the water.

2 For each of the filters described above, identify the **residue** and the **filtrate**. The first one has been done for you.

Filter	Residue	Filtrate
gas masks	poisonous gases and chemicals	clean air
tea bags		
air conditioners		
water jugs		
coffee machines		
oil filters		
clothes dryers		

Evaporation and crystallisation

Filtration cannot be used to separate substances that have been dissolved in a solvent, because the particles are too small and pass through the filter paper. Heat can be used to separate a liquid from a solution, leaving the solute behind. This process is called **evaporation**.

 **1** Circle the correct option from each pair in brackets.

Filtration is used to remove (soluble/insoluble) substances from a liquid.

Evaporation is used to remove (soluble/insoluble) substances from a liquid.

When you separate a solution using evaporation, you (keep/lose) the solvent.

When you separate a solution using evaporation, you are left with the (solute/solvent).

Evaporating the water from a solution by heating is a quick method of separating the solute from the solvent. However, sometimes evaporation is not a suitable method because the quantities are too large, or you want large crystals. In these cases, **crystallisation** can be used instead. For example, in the production of commercial salt, the sun's heat is used to evaporate the water. Large salt crystals form because the water is evaporated slowly.

2 Decide whether each of the following statements is true or false.

Evaporation allows you to keep the separated solvent and solute. _____

Filtration can't be used to separate the solute and solvent from a solution. _____

If you evaporate the liquid quickly, the crystals formed are small. _____

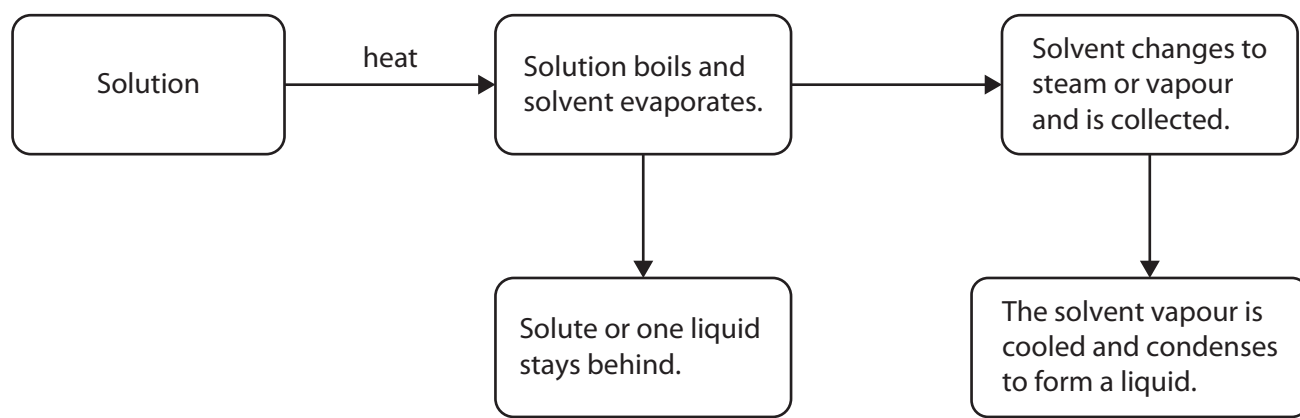
Crystallisation occurs when a solution is heated quickly. _____

Filtration is used to remove insoluble substances from a liquid. _____

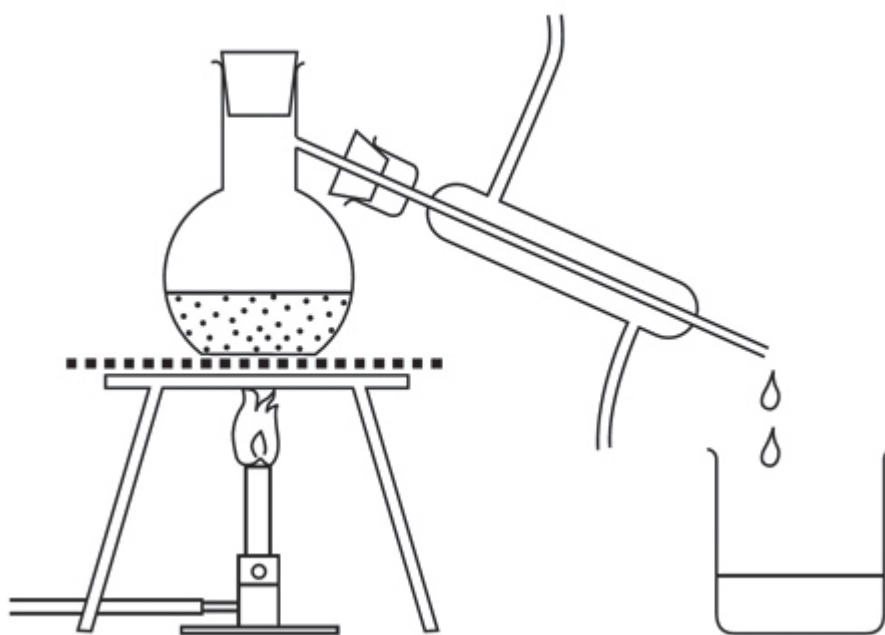
Distillation (1)

Distillation is used to separate dissolved substances when you want to keep both the liquid and the solute. Distillation can also be used to separate a liquid from another liquid. This is because different liquids have different boiling points. The process of distillation uses two changes of state, **evaporation** and **condensation**. The material collected during distillation is called the **distillate**.

The flowchart below shows the steps involved in **distillation**.



Distillation is used to make alcoholic spirits, to produce oils and perfumes from plants, to make distilled water and to refine oil. In a science laboratory a **Leibig's Condenser** can be used for distillation. The condenser uses cold water in a tube to cool the evaporated liquid.



When set up, a Leibig's Condenser looks like this.

Distillation (2)

Read the Resource sheet 'Distillation (1)', then answer the following questions.

 1 How is distillation different to evaporation?

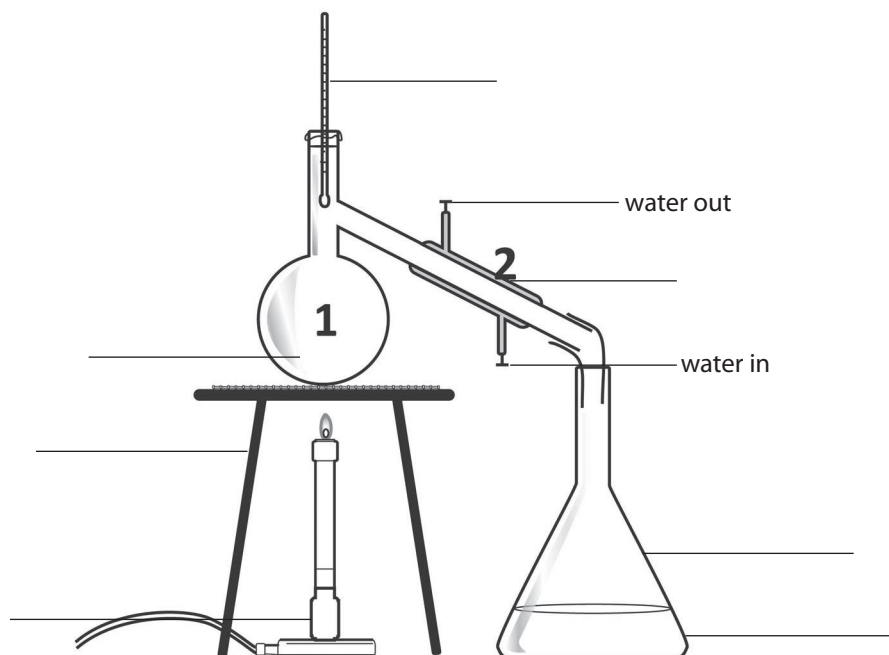
2 Which two changes of state are involved in distillation?

3 Jeremy wants to separate and collect the solvent from a salt solution using the process of distillation. Number the steps he needs to take in the correct order.

_____ Gently heat the salt solution in the conical flask until nearly all the water has evaporated.

_____ Set up equipment as shown in diagram below.

_____ Connect tubing to Leibig's Condenser and turn on water. Run water through condenser.



Word bank

salt water
thermometer
tripod
conical flask
Bunsen burner
pure water
condenser

4 Use the words in the word bank to label the diagram.

5 Name the change of state that occurs at:

1 _____

2 _____

Objectives

- To identify the differences between chemical and physical changes
- To recognise that in chemical changes, new substances are formed
- To understand that chemical changes cannot be easily reversed
- To identify evidence that a chemical change has occurred
- To develop students' ability to safely conduct experiments and to record and analyse data

Prior knowledge

Students should understand that changes to materials can be reversible or irreversible. Reversible changes, such as change of state, can be turned back easily, but irreversible changes cannot.

Background

A physical change occurs when matter changes but no new substance forms. A physical change may involve a change of state, a substance dissolving, or changes in shape or size. Physical changes can usually be reversed. A chemical change occurs when a new substance with different properties is formed during a chemical reaction. A chemical change is irreversible.

Starter activity

Odd one out: Show students three different images (melting, dissolving and a chemical reaction), and ask them to pick the odd one out. Brainstorm reasons why that is the odd one out. Video clips showing three different changes to matter can be found on YouTube.

Resource and Activity sheets

After reading the Resource sheet 'Changing matter' and completing the Activity sheet 'Physical and chemical changes' students should be able to explain the difference between physical and chemical changes and identify examples of each.

The Activity sheet 'Is it a chemical change?' develops students' skills in safely conducting experiments and recording and analysing data. Students need to look for evidence of a chemical change while observing different reactions. Although most chemicals used are harmless, ensure safety glasses are worn at all times

to ensure that ammonia and bleach cannot get in students' eyes.

The 'Evidence of a chemical change' Activity sheet encourages students to identify chemical reactions. Students are given some observations about different experiments that involved a chemical change. They read the descriptions and identify the evidence used to determine that a chemical change occurred in each case.

The 'Precipitates' Activity sheet explains that a precipitate is a solid that forms when two liquids are mixed together. Students conduct an experiment to find out which liquids form a precipitate when mixed together.

Extension ideas

- Students could produce a concept map showing chemical and physical changes.
- Students could develop a 'Jeopardy' game for properties and changes in matter.

Selected answers

Physical and chemical changes 2 physical; chemical – heat and light produced, new substance (ash); chemical – new substance; chemical – gas produced, heat produced; physical; physical



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Changing matter

Changes to matter can be classified as either **physical** or **chemical** changes. During a **physical change** some of the characteristics of the substance change, but the substance remains the same. No new substance is formed.

When we boil water, it changes into steam. Steam is another form of water. When water is heated, this does not create a new substance – it simply changes the water's state from a liquid to a gas. Examples of physical changes to matter include:

- changes of state (solid, liquid, gas)
- changes in shape or size
- dissolving.

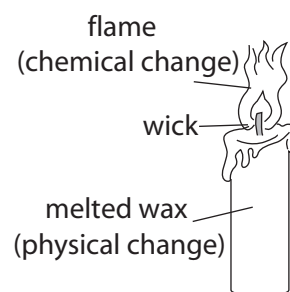


Physical changes are usually easy to change back, or **reverse**. For example, if you dissolve salt in water, you can separate the salt back out by boiling the water.

When a **chemical change** occurs, the original substance is changed into something new. A chemical change is caused by a chemical reaction. A chemical change is **irreversible** – you cannot get the original substance back. For example, when a candle is lit, some of the wax melts, which is a physical change, and some of it burns and changes to ash and smoke, which is a chemical change.

To identify that a chemical change has taken place, we can look for the following clues:

- The substance fizzes or bubbles (gas is produced).
- Smoke is produced.
- Heat is produced.
- Light energy is produced (the substance glows, or a flame is produced).
- You cannot get the original substance back.
- There is a permanent colour change.
- There is a smell.
- A solid forms when two liquids are mixed.



Physical and chemical changes

Read the Resource sheet 'Changing matter', then answer the following questions.

-  **1** Unjumble the sentence parts and then rewrite each sentence about physical and chemical changes in the correct order.

are physical changes. that do not produce Changes in matter new substances

is an A change of state physical change. example of a

reversible. changes are Most physical

is an example Burning the wick of a candle chemical change. of a

new substance change occurs a When a chemical is formed.

- 2** Identify which changes in the table below are physical changes and which are chemical changes. For chemical changes, complete the clue that shows it is a chemical change. The first one has been done for you.

Change to matter	Physical or chemical change	Clue that shows it is a chemical change
fireworks going off	chemical change	heat and light produced
water freezing		
lighting a match		
cooking a cake		
baking soda in vinegar		
crushing a piece of chalk		
melting chocolate		

Is it a chemical change?



The following experiment tests for evidence of a chemical change.

Experiment: Which are chemical changes?

Aim: To find out which substances react to produce a chemical change when mixed together.

Equipment

- 4 test tubes
- dropper bottles of vinegar, bleach, lemon juice and ammonia
- test-tube rack
- spatula
- safety glasses (**Ensure safety glasses are worn at all times.**)

Method

- 1 Place 1 cm of baking soda in each test tube.
- 2 Place 10 drops of vinegar in test tube 1. Record your observations in the table below.
- 3 Place 10 drops of bleach in test tube 2. Record your observations.
- 4 Place 10 drops of lemon juice in test tube 3. Record your observations.
- 5 Place 10 drops of ammonia in test tube 4. Record your observations.

 **1** Complete the experiment and fill in the table below, placing ticks in the boxes.

Evidence of change	Vinegar	Bleach	Lemon juice	Ammonia
gas (bubbles)				
light or sound				
heat (feel test tube)				
smell				

2 Which substances combined to produced a chemical change?

3 Was it hard to decide whether some changes were physical or chemical? Why?

Evidence of a chemical change

-  1 Read the following descriptions of experiments that involved a chemical change. Underline the evidence of a chemical change in each experiment.

Experiment A

A test tube half full of green copper carbonate solution was heated using a Bunsen burner. The solution turned black and bubbles were produced.

Experiment B

A piece of magnesium ribbon was placed in a test tube of sulfuric acid. The liquid started to fizz and after a while the test tube felt hot to touch. When the reaction stopped, the piece of magnesium was no longer in the acid.

Experiment C

Silver nitrate solution and sodium chloride solution are both clear and colourless. When silver nitrate is poured into a test tube of sodium chloride, a white solid forms and sinks to the bottom of the test tube.

Experiment D

A spoonful of magnesium carbonate powder is mixed in a small beaker of sulfuric acid. The acid starts to bubble. When touched, the beaker feels warm.

Experiment E

The teacher drops a small piece of potassium into a beaker of water. The potassium bursts into flames and starts to move around on top of the water. It makes a sizzling sound and then the flame goes out. Soon, a horrible smell is detected. The beaker is warm to touch when the teacher pours out the water.

Experiment F

When a piece of magnesium is placed into the flame of a Bunsen burner it catches alight and emits a bright light. Once it has burnt out, all that is left is a fine grey powdery substance.

Precipitates



A **precipitate** is a solid that forms when two liquids are mixed together. The formation of a precipitate is evidence of a chemical change.

Experiment: Forming a precipitate

You are now going to conduct an experiment to see which chemicals form a precipitate when mixed together.

Equipment

- 5 test tubes
- test-tube rack
- safety glasses
- Dropper bottles of silver nitrate, sodium chloride, sodium hydroxide, copper sulfate, sodium carbonate

Method

- 1 Put on safety glasses.
- 2 Collect 5 test tubes and place in a test-tube rack.
- 3 In each test tube, mix combinations of the chemicals listed in the table below.

 1 Perform the experiment and record your results in the table below.

Chemicals mixed	Observations (draw what it looked like)	Did a precipitate form?
Silver nitrate and sodium chloride		
Sodium chloride and sodium hydroxide		
Copper sulfate and sodium hydroxide		
Copper sulfate and sodium carbonate		

Objectives

- To identify the reactants and products in a chemical reaction
- To use word equations to represent simple reactions
- To understand common chemical reactions such as rusting and burning and use word equations to describe them
- To recognise that photosynthesis and respiration are chemical reactions that occur in nature

Prior knowledge

Students should understand that changes to matter can be physical or chemical. A chemical change occurs when a new substance is formed.

Background

When a chemical reaction occurs, new substances are formed which are different from the original substances. The original substances are called the reactants, and the new substances are called the products. We use word equations to represent chemical reactions. The reactants are shown on the left of the equation and the products on the right.

Starter activity

Demonstrate the reaction between table sugar and sulfuric acid. Ask students to compare the properties of the original substances to the properties of the substance you end up with. Discuss the evidence that tells us that a chemical change occurred. Finally, introduce students to the concept of word equations by displaying several examples and asking students to explain what each represents.

Activity sheets

The Activity sheet 'Reactants and products' introduces students to word equations. Students read a short passage and then rewrite incorrect statements to make them correct. Students then identify the reactants and products in a simple word equation. The 'Writing word equations' Activity sheet provides students with practice writing word equations. Students identify the reactants and products from written sentences and then write word equations.

The Activity sheet 'Rusting' introduces this common chemical reaction. Students summarise notes about this process to demonstrate their understanding. The lesson concludes with some higher-order questions.

The Activity sheet 'Burning' explains the chemical reaction of burning (combustion) and the effects of burning fossil fuels on the environment. Students complete a cloze passage with the help of the word bank, then consolidate their understanding by answering questions related to the passage. Students also complete a matching phrases activity.

'Chemical reactions in nature' explains that photosynthesis and respiration are two chemical reactions that occur in nature. Students read the passage provided and the word equations for photosynthesis and respiration. They then complete a crossword to consolidate understanding.

Extension ideas

- Students could investigate the effect of temperature on reaction rate.
- Provide students with word equation cards which include reactants, plus signs, arrows and products. Have students form word equations using the cards. This is a good kinaesthetic activity.
- Students could investigate how to get the largest 'pop' from burning hydrogen.

Selected answers

Burning 4 methane + oxygen → carbon dioxide + water

Chemical reactions in nature 2 Across:

3 photosynthesis **4** energy **6** respire **7** chlorophyll

8 sunlight **Down:** **1** water **2** oxygen **3** products

5 equations



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Reactants and products

When a **chemical reaction** occurs, new substances are made. The substances that we start with are called the **reactants**. The new substances that are made are called the **products**. The properties of the products are different from the properties of the reactants.

We can describe what happens in a chemical reaction by writing a **word equation**.



The arrow means 'changes into' or 'goes to'. It also shows you the direction of the reaction, so that you do not mix up the reactants and products.

-  **1** What do we call the substances that we start with? _____
- 2** What do we call the new substances that form after a chemical reaction has occurred?

- 3** How can we describe what is happening in a chemical reaction?

- 4** Rewrite each of the following statements to make it correct.
 - a** The reactants are written on the right-hand side of the arrow in a word equation.

 - b** The products are the substances that we started with before the chemical reaction occurred.

 - c** We use an equals sign to represent 'goes to' when writing word equations.

When hydrogen reacts with oxygen, water is formed. This reaction can be shown with the following word equation:



- 5** Underline the reactant/s in red and product/s in blue.

Writing word equations

When carbon is heated it reacts with oxygen to form the product carbon dioxide. The two reactants, carbon and oxygen, are elements that form the compound carbon dioxide. This chemical reaction can be shown in the following **word equation**.



We write 'heat' above the arrow to show that heat is needed for the reaction to occur.

 **1** For each of the following reactions:

- underline the reactants in red and the products in blue.
 - write a word equation showing the reaction.
- a** When you mix the elements sulphur and oxygen together and heat them, the compound sulfur dioxide is produced.

- b** Magnesium reacts with bromine to produce magnesium bromide.

- c** When zinc is placed in hydrochloric acid it reacts, producing bubbles of hydrogen gas and a new substance called zinc chloride.

- d** When iron reacts with oxygen it forms rust. The chemical name for rust is iron oxide.

- e** Sodium reacts with chlorine to form sodium chloride.

- 2** Use the information in the following word equation to complete the passage below.



When copper _____ is placed in _____ acid it undergoes a chemical reaction. Two new substances are produced: copper _____ and _____.

Rusting

- Chemical reaction between iron and oxygen
- Produces a new substance – iron oxide
- Rusting occurs when iron or steel is left in damp conditions
- Paint protects the iron from water and oxygen
- Salt speeds up rusting
- Rust = reddish brown flakes
- Things become weak and break more easily

 1 Use the notes above to write a paragraph about rusting.

2 What are the reactants in this chemical reaction?

_____ and _____

3 What is the product? _____

4 Write the word equation for rusting.

5 What is the problem with objects that have rusted?

6 What is one way that we can prevent rusting?

7 Why do objects near the sea rust more quickly?

Burning

 1 Use the words in the word bank to complete the passage below.

Burning is a c_____ reaction that occurs when a substance combines with oxygen and produces h_____ and light energy. Another name for burning is c_____. F_____ are substances that release e_____ when they burn. Gas and o_____ are two examples of fuels. When they b_____ they produce carbon d_____ and w_____. The word equation for the burning of oil is:

oil + o_____ → c_____ dioxide + water.

Carbon dioxide is a g_____ gas. Burning f_____ fuels such as oil, c_____ and natural gas releases carbon dioxide into the a_____. Greenhouse gases trap heat in the atmosphere, which makes the earth h_____. This is called global w_____.

2 What is needed for a fuel to burn? _____

3 What two products are produced when fuels are burnt? _____

4 Natural gas is made of a compound called methane. Write a word equation for the combustion of natural gas.

m_____ + _____ → _____ + _____

5 Draw lines matching the sentence halves to complete the statements below.

Oil, coal and natural gas are

is global warming.

Carbon dioxide is

releases carbon dioxide.

Burning fossil fuels

fossil fuels.

Greenhouse gases trap

a greenhouse gas.

The scientific name for the earth heating up

heat in the atmosphere.

Word bank

atmosphere
heat
fossil
Fuels
warming
combustion
dioxide
hotter
chemical
oil
oxygen
greenhouse
coal
energy
burn
water
carbon

Chemical reactions in nature

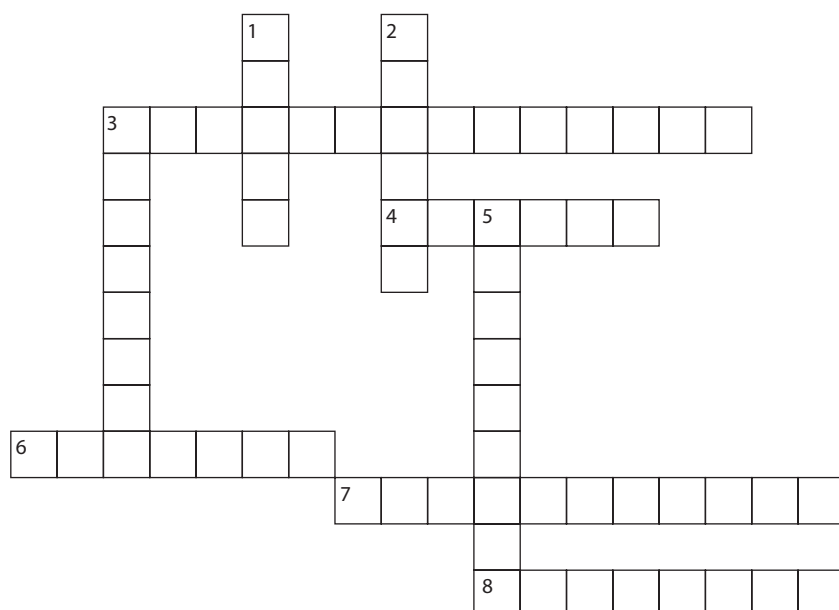
Two chemical reactions that occur in nature are **photosynthesis** and **respiration**.

Photosynthesis is the process by which green plants use carbon dioxide and water to produce food (glucose) and oxygen. Respiration is the process by which living things use food (glucose) and oxygen to produce energy. Carbon dioxide and water are also produced. All living things respire. The word equations for these reactions are:

Photosynthesis: carbon dioxide + water $\longrightarrow$ glucose + oxygen

Respiration: glucose + oxygen $\longrightarrow$ carbon dioxide + water + energy
released

 **1** Use the words in the word bank to complete the crossword puzzle.



Word bank

respire
oxygen
equations
photosynthesis
sunlight
water
energy
chlorophyll
products

Across

- 3** The process by which green plants make food
- 4** Released when respiration occurs
- 6** All living things do this
- 7** Found in the leaves of green plants
- 8** Needed for photosynthesis to occur

Down

- 1** One of the reactants of photosynthesis
- 2** Glucose and _____ are the reactants for respiration.
- 3** Glucose and water are the _____ of photosynthesis.
- 5** Chemical reactions can be written as word _____.

Your *Achieve!* Science understanding CD-ROM

User agreement

© Janine Haeusler and Blake Education 2014

- You may install this CD-ROM onto a server for access by students at one school campus only.
- You may not otherwise copy, print, store or transmit any part of this CD-ROM unless you have written permission from Blake Education or a copyright exception applies under the Copyright Act, 1968. For inquiries and permissions please contact:
Jennifer Gough
Blake Education
655 Parramatta Road, Leichhardt, NSW, 2040.
Phone: 02 8585 4058
- The attached CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. Do not break this seal or remove the CD-ROM unless you have purchased this book.
- Unless the CD-ROM is faulty, this book cannot be returned from preview, or credit or refund provided, if the CD-ROM wallet seal is broken.
- No part of the book or CD-ROM can be copied and distributed to, or used by, a non-purchasing school campus or institution. A campus is defined as a school that may be a part of a larger multi-campus institution.
- Purchasers do not have the right to resell, distribute, or pass off as their own, any still images, text, diagrams or software contained in the book or CD-ROM.
- All photocopies must bear the Blake Education copyright line.
- Purchasers may not remove or obscure the publisher copyright and trademark notices from the book or CD-ROM.
- The purchaser is entirely responsible for their computer hardware and system software and for ensuring that they are able to correctly operate the CD-ROM and its associated software.

Copying this disc isn't just illegal – it affects an independent, Australian-owned and operated business and the people who work for it. Both individuals and institutions are liable for large fines for breaking copyright laws.

Thank you for your cooperation!

Achieve! Instant Lessons

Quality ready-to-use resources

Each book in the *Achieve!* series has been designed and written for **secondary students who have low-level literacy skills** and require **modified** classroom activities to fully participate in the curriculum. Students will **experience success** as they work alongside their classmates on the same subjects with material that is tailored for their literacy needs.

Key features

- ✓ Mature-look page layout and illustrations
- ✓ Topics that will interest teenagers
- ✓ An emphasis on literacy skills
- ✓ Clear learning objectives
- ✓ Starter activities, background topic information and ideas for summarising a lesson

The work sheets in this series are ideal for group or individual study and are easy for classroom helpers to implement in one-on-one lessons. You can also use the photocopiable pages for revision, assessment and home study.

Achieve! Science understanding CHEMISTRY FOR YEARS 7–8

While very suitable for the regular classroom, each of the *Achieve!* Science titles has been written at a literacy level that is appropriate for secondary school students who, because of poor reading skills, may struggle with regular classroom materials.

This title will assist teachers to develop lessons for the chemical sciences sub-strand in *Science understanding* for years 7 and 8. Units of work cover a variety of topics, including the different states of matter, elements, compounds and mixtures, solutions and a range of separation techniques and physical and chemical changes. Students build their scientific vocabulary and understanding of key concepts. Work sheets involving practical activities also develop the ability to plan and safely conduct experiments and to record and analyse data.

The CD-ROM contains all the pages that are featured in the book, as well as a file of selected illustrations that can be used for other work sheets. These files can be easily edited to suit the needs of students.

Topics include

Matter
Solids, liquids and gases
Changes of state
Atoms and elements
Metals and non-metals

Compounds
Mixtures and solutions
Separating mixtures
Physical and chemical changes
Chemical reactions

Associated titles in the Achieve! Instant Lessons series

Achieve! Science series

- Science understanding
 - Chemistry for years 9–10
 - Biology for years 7–8
 - Biology for years 9–10
- Science as a human endeavour
 - Biology for years 7–8
 - Biology for years 9–10
- Indigenous contexts
- Sustainability in the Asia Pacific
- Literacy for science
- Measurement and laboratory skills

Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Rd Clayton South VIC 3169
+61 3 9558 4433
www.blake.com.au



Made for the
Australian Curriculum

ISBN: 978-1-92571-061-8



9 781925 710618